Assignment 1 - Data Cleaning - Instructions

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04/09/2020

Assignment 1, Language development in Autism Spectrum Disorder (ASD) - Brushing up your code skills

In this first part of the assignment we will brush up your programming skills, and make you familiar with the data sets you will be analysing for the next parts of the assignment.

In this warm-up assignment you will: 1) Create a Github (or gitlab) account, link it to your RStudio, and create a new repository/project 2) Use small nifty lines of code to transform several data sets into just one. The final data set will contain only the variables that are needed for the analysis in the next parts of the assignment 3) Warm up your tidyverse skills (especially the sub-packages stringr and dplyr), which you will find handy for later assignments.

Learning objectives:

- Become comfortable with tidyverse (and R in general)
- Test out the git integration with RStudio
- Build expertise in data wrangling (which will be used in future assignments)

0. First an introduction on the data

Language development in Autism Spectrum Disorder (ASD)

Reference to the study: https://www.ncbi.nlm.nih.gov/pubmed/30396129

Background: Autism Spectrum Disorder (ASD) is often related to language impairment, and language impairment strongly affects the patients ability to function socially (maintaining a social network, thriving at work, etc.). It is therefore crucial to understand how language abilities develop in children with ASD, and which factors affect them (to figure out e.g. how a child will develop in the future and whether there is a need for language therapy). However, language impairment is always quantified by relying on the parent, teacher or clinician subjective judgment of the child, and measured very sparcely (e.g. at 3 years of age and again at 6).

In this study the researchers videotaped circa 30 kids with ASD and circa 30 comparison kids (matched by linguistic performance at visit 1) for ca. 30 minutes of naturalistic interactions with a parent. They repeated the data collection 6 times per kid, with 4 months between each visit. Then the researchers transcribed the data and counted: i) the amount of words that each kid uses in each video. Same for the parent. ii) the amount of unique words that each kid uses in each video. Same for the parent. iii) the amount of morphemes per utterance (Mean Length of Utterance) displayed by each child in each video. Same for the parent.

Different researchers involved in the project provide you with different datasets: 1) demographic and clinical data about the children (recorded by a clinical psychologist) 2) length of utterance data (calculated by a linguist) 3) amount of unique and total words used (calculated by a jack-of-all-trade, let's call him RF)

Your job in this assignment is to double check the data and make sure that it is ready for the analysis proper (Assignment 2), in which we will try to understand how the children's language develops as they grow as a function of cognitive and social factors and which are the "cues" suggesting a likely future language impairment.

1. Let's get started on GitHub

In the assignments you will be asked to upload your code on Github and the GitHub repositories will be part of the portfolio, therefore all students must make an account and link it to their RStudio (you'll thank us later for this!).

Follow the link to one of the tutorials indicated in the syllabus: * Recommended: https://happygitwithr.com/ * Alternative (if the previous doesn't work): https://support.rstudio.com/hc/en-us/articles/200532077-Version-Control-with-Git-and-SVN * Alternative (if the previous doesn't work): R.version.string N.B. Create a GitHub repository for the Assignment 1 and link it to a project on your RStudio.

2. Now let's take dirty dirty data sets and make them into a tidy one

If you're not in a project in Rstudio, make sure to set your working directory here. If you created an RStudio project, then your working directory (the directory with your data and code for these assignments) is the project directory.

```
pacman::p_load(tidyverse,janitor)
```

Load the three data sets, after downloading them from dropbox and saving them in your working directory: * Demographic data for the participants: https://www.dropbox.com/s/lol8f5m4pgrpmto/demo_train.csv?dl=0 * Length of utterance data: https://www.dropbox.com/s/usyauqm37a76of6/LU_train.csv?dl=0 * Word data: https://www.dropbox.com/s/8ng1civpl2aux58/token_train.csv?dl=0

```
#ANDERS
#Loading data sets
demo_train <- read.csv("demo_train.csv")
LU_train <- read.csv("LU_train.csv")
token_train <- read.csv("token_train.csv")

#Checking how many participants in each df
length(unique(demo_train$Child.ID))
```

```
## [1] 67
```

```
length(unique(LU_train$SUBJ))
```

```
## [1] 61
```

```
length(unique(token_train$SUBJ))
```

```
## [1] 61
```

```
#Investigating how many times each child participated
demo_train %>% group_by(Child.ID) %>% summarise(length(Child.ID))
```

```
## `summarise()` ungrouping output (override with `.groups` argument)
```

```
## # A tibble: 67 x 2
##    Child.ID `length(Child.ID)`
##    <fct>                 <int>
##  1 A.A.                      1
##  2 A.D.                      6
##  3 A.H.                      6
##  4 A.R.                      6
##  5 A.S.                      6
##  6 A.Z.                      6
##  7 Adam                      6
##  8 Albert                    6
##  9 Alfie                     6
## 10 Allison                   6
## # ... with 57 more rows
```

```
LU_train %>% group_by(SUBJ) %>% summarise(length(SUBJ))
```

```
## `summarise()` ungrouping output (override with `.groups` argument)
```

```
## # A tibble: 61 x 2
##    SUBJ     `length(SUBJ)`
##    <fct>             <int>
##  1 A.D.                  6
##  2 A.H.                  6
##  3 A.R.                  5
##  4 A.S.                  5
##  5 A.Z.                  6
##  6 Adam.                 5
##  7 Albert.               6
##  8 Alfie.                6
##  9 Allison.              6
## 10 Annie.                5
## # ... with 51 more rows
```

```
token_train %>% group_by(SUBJ) %>% summarise(length(SUBJ))
```

```
## `summarise()` ungrouping output (override with `.groups` argument)
```

```
## # A tibble: 61 x 2
##    SUBJ     `length(SUBJ)`
##    <fct>             <int>
##  1 A.D.                  6
##  2 A.H.                  6
##  3 A.R.                  5
##  4 A.S.                  5
##  5 A.Z.                  6
##  6 Adam.                 5
##  7 Albert.               6
##  8 Alfie.                6
##  9 Allison.              6
## 10 Annie.                5
## # ... with 51 more rows
```

Explore the 3 datasets (e.g. visualize them, summarize them, etc.). You will see that the data is messy, since the psychologist collected the demographic data, the linguist analyzed the length of utterance in May 2014 and the fumbling jack-of-all-trades analyzed the words several months later. In particular: - the same variables might have different names (e.g. participant and visit identifiers) - the same variables might report the values in different ways (e.g. participant and visit IDs) Welcome to real world of messy data :-)

Before being able to combine the data sets we need to make sure the relevant variables have the same names and the same kind of values.

So:

2a. Identify which variable names do not match (that is are spelled differently) and find a way to transform variable names. Pay particular attention to the variables indicating participant and visit.

Tip: look through the chapter on data transformation in R for data science (http://r4ds.had.co.nz). Alternatively you can look into the package dplyr (part of tidyverse), or google "how to rename variables in R". Or check the janitor R package. There are always multiple ways of solving any problem and no absolute best method.

```
#SIGRID
#Changing collumn names
colnames(demo_train)[1:2] <- c("SUBJ","VISIT")
```

2b. Find a way to homogeneize the way "visit" is reported (visit1 vs. 1).

Tip: The stringr package is what you need. str_extract () will allow you to extract only the digit (number) from a string, by using the regular expression \d.

```
#GUSTAV
#Changing visit number to numeric
LU_train$VISIT <- str_extract(LU_train$VISIT, "\\d")
token_train$VISIT <- str_extract(token_train$VISIT, "\\d")

#Changing visit variable into a factor:
demo_train$VISIT <- as.factor(demo_train$VISIT)
LU_train$VISIT <- as.factor(LU_train$VISIT)
token_train$VISIT <- as.factor(token_train$VISIT)
```

2c. We also need to make a small adjustment to the content of the Child.ID coloumn in the demographic data. Within this column, names that are not abbreviations do not end with "." (i.e. Adam), which is the case in the other two data sets (i.e. Adam.). If The content of the two variables isn't identical the rows will not be merged. A neat way to solve the problem is simply to remove all "." in all datasets.

Tip: stringr is helpful again. Look up str_replace_all Tip: You can either have one line of code for each child name that is to be changed (easier, more typing) or specify the pattern that you want to match (more complicated: look up "regular expressions", but less typing)

```
#SIGRID
#Changing subject.ID.
demo_train$SUBJ <- str_replace_all(demo_train$SUBJ, "[[:punct:]]", "")
LU_train$SUBJ <- str_replace_all(LU_train$SUBJ, "[[:punct:]]", "")
token_train$SUBJ <- str_replace_all(token_train$SUBJ, "[[:punct:]]", "")

#Making a factor
demo_train$SUBJ <- as.factor(demo_train$SUBJ)
LU_train$SUBJ <- as.factor(LU_train$SUBJ)
token_train$SUBJ <- as.factor(token_train$SUBJ)
```

2d. Now that the nitty gritty details of the different data sets are fixed, we want to make a subset of each data set only containig the variables that we wish to use in the final data set. For this we use the tidyverse package dplyr, which contains the function select().

The variables we need are: * Child.ID, * Visit, * Diagnosis, * Ethnicity, * Gender, * Age, * ADOS, * MullenRaw, * ExpressiveLangRaw, * Socialization * MOT_MLU, * CHI_MLU, * types_MOT, * types_CHI, * tokens_MOT, * tokens_CHI.

Most variables should make sense, here the less intuitive ones. * ADOS (Autism Diagnostic Observation Schedule) indicates the severity of the autistic symptoms (the higher the score, the worse the symptoms). Ref: https://link.springer.com/article/10.1023/A:1005592401947 * MLU stands for mean length of utterance (usually a proxy for syntactic complexity) * types stands for unique words (e.g. even if "doggie" is used 100 times it only counts for 1) * tokens stands for overall amount of words (if "doggie" is used 100 times it counts for 100) * MullenRaw indicates non verbal IQ, as measured by Mullen Scales of Early Learning (MSEL https://link.springer.com/referenceworkentry/10.1007%2F978-1-4419-1698-3_596) * ExpressiveLangRaw indicates verbal IQ, as measured by MSEL * Socialization indicates social interaction skills and social responsiveness, as measured by Vineland (https://cloudfront.ualberta.ca/-/media/ualberta/faculties-and-programs/centres-institutes/community-university-partnership/resources/tools---assessment/vinelandjune-2012.pdf)

Feel free to rename the variables into something you can remember (i.e. nonVerbalIQ, verbalIQ)

```
#ANDERS
#Selecting relevant variables for each df
variables_to_include1 <- c("SUBJ", "VISIT", "Diagnosis","Ethnicity", "Gender", "Age", "ADOS",  "MullenRa
variables_to_include2 <- c("SUBJ", "VISIT", "MOT_MLU", "CHI_MLU")
variables_to_include3 <- c("SUBJ", "VISIT", "types_MOT", "types_CHI", "tokens_MOT", "tokens_CHI")

#subsetting relevant variables
demo_train <- select(demo_train, variables_to_include1)
```

```
## Note: Using an external vector in selections is ambiguous.
## i Use ‘all_of(variables_to_include1)‘ instead of ‘variables_to_include1‘ to silence this message.
## i See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This message is displayed once per session.
```

```
LU_train <- select(LU_train, variables_to_include2)
```

```
## Note: Using an external vector in selections is ambiguous.
## i Use ‘all_of(variables_to_include2)‘ instead of ‘variables_to_include2‘ to silence this message.
## i See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This message is displayed once per session.
```

```
token_train <- select(token_train, variables_to_include3)
```

```
## Note: Using an external vector in selections is ambiguous.
## i Use ‘all_of(variables_to_include3)‘ instead of ‘variables_to_include3‘ to silence this message.
## i See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This message is displayed once per session.
```

2e. Finally we are ready to merge all the data sets into just one.

Some things to pay attention to: * make sure to check that the merge has included all relevant data (e.g. by comparing the number of rows) * make sure to understand whether (and if so why) there are NAs in the dataset (e.g. some measures were not taken at all visits, some recordings were lost or permission to use was withdrawn)

```r
#GUSTAV
#Merging the dataframes:
df <- merge(demo_train, LU_train, by = c("SUBJ","VISIT"), all = T)
df <- merge(df, token_train, by = c("SUBJ","VISIT"), all = T)
```

2f. Only using clinical measures from Visit 1 In order for our models to be useful, we want to miimize the need to actually test children as they develop. In other words, we would like to be able to understand and predict the children's linguistic development after only having tested them once. Therefore we need to make sure that our ADOS, MullenRaw, ExpressiveLangRaw and Socialization variables are reporting (for all visits) only the scores from visit 1.

A possible way to do so: * create a new dataset with only visit 1, child id and the 4 relevant clinical variables to be merged with the old dataset * rename the clinical variables (e.g. ADOS to ADOS1) and remove the visit (so that the new clinical variables are reported for all 6 visits) * merge the new dataset with the old

```r
#SIGRID
#Creating new df for visit 1 only including the relevant variables.
visit1 <- select(filter(df, df$VISIT==1), c("SUBJ","VISIT","ADOS", "MullenRaw","ExpressiveLangRaw","Soc

#Renaming the collumns:
new_names <- c("ADOS1", "MullenRaw1","ExpressiveLangRaw1","Socialization1")
colnames(visit1)[3:6] <- new_names

#Removing visit variable and merging
visit1$VISIT <- NULL
df <- merge(df,visit1, by = "SUBJ")
```

2g. Final touches

Now we want to * anonymize our participants (they are real children!). * make sure the variables have sensible values. E.g. right now gender is marked 1 and 2, but in two weeks you will not be able to remember, which gender were connected to which number, so change the values from 1 and 2 to F and M in the gender variable. For the same reason, you should also change the values of Diagnosis from A and B to ASD (autism spectrum disorder) and TD (typically developing). Tip: Try taking a look at ifelse(), or google "how to rename levels in R". * Save the data set using into a csv file. Hint: look into write.csv()

```r
#ANDERS
#Anomymizing the names
df$SUBJ <- as.integer(df$SUBJ)
df$SUBJ <- as.factor(df$SUBJ)

#Changing gender and diagnosis names:
df$Gender <- as.factor(df$Gender)
levels(df$Gender) <- c("M","F")
levels(df$Diagnosis) <- c("ASD","TD")
#...
```

```r
#GUSTAV
#Saving the df as a CSV
write_csv(df,path="ASD_Children.csv")
```

3) BONUS QUESTIONS The aim of this last section is to make sure you are fully fluent in the tidyverse. Here's the link to a very helpful book, which explains each function: http://r4ds.had.co.nz/index.html

4) USING FILTER List all kids who:

1. have a mean length of utterance (across all visits) of more than 2.7 morphemes.
2. have a mean length of utterance of less than 1.5 morphemes at the first visit
3. have not completed all trials. Tip: Use pipes to solve this

USING ARRANGE

1. Sort kids to find the kid who produced the most words on the 6th visit
2. Sort kids to find the kid who produced the least amount of words on the 1st visit.

USING SELECT

1. Make a subset of the data including only kids with ASD, mlu and word tokens
2. What happens if you include the name of a variable multiple times in a select() call?

USING MUTATE, SUMMARISE and PIPES 1. Add a column to the data set that represents the mean number of words spoken during all visits. 2. Use the summarise function and pipes to add an column in the data set containing the mean amount of words produced by each trial across all visits. HINT: group by Child.ID 3. The solution to task above enables us to assess the average amount of words produced by each child. Why don't we just use these average values to describe the language production of the children? What is the advantage of keeping all the data?

Assignment 2

Study group: ANDERS WHEELMAN, GUSTAV HELMET AND SIGRID SNAPFIELD.
Due 28/9

Assignment 2 part 1: Basic description of language development

Study group: Anders, Gustav H & Sigrid
Due 16/9
Link to code: https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_2
The file is called 'FINAL_A2_P1_LanguageASD_instructions.Rmd'

1. **Describe your sample (n, age, gender, clinical and cognitive features of the two groups) and critically assess whether the groups (ASD and TD) are balanced**

Descriptive statistics for the overall sample

(Anders) The groups are relatively balanced according to ethnicity (the study was executed in Denmark), non-verbal IQ and number of children in each group. The participants were, however, mostly male.
Overall assessment of the data:

- n = 372 (ASD = 176, TD = 196)
- men > women in both groups (ADS: 150 M, 26 F, TD: 160 M, 36 F)
- ethnicity = white > other ethnicities (ASD: 130 white kids, other minorities: 46, TD: 190 white kids, other minorities: 6)
- Non-verbal IQ: ASD mean 33.15, TD mean 35.8

We see clear differences between ASD and TD when it comes to measures of verbal IQ and ADOS. Across all visits we also see the following difference in their linguistic capacities.

- We see a clear difference in autistic measure, ADOS (mean of ADS = 13.8, mean of TD = 1.2)
- Verbal IQ: ASD mean 22.62, TD mean: 29.29
- Mean MLU for the children: ASD: 1.6, TD: 2.3

- Mean Types (unique words used): ASD: 77, TD: 129
- Mean Tokens (amount of words used): ASD: 293, TD: 475

Descriptive statistics for visit 1

(Gustav) The ASD children were matched with TD children by their linguistic skills. This can be seen by the descriptive statistics of the children's linguistic skills at their first visit. This entails that the age difference between the two groups is approximately 1 year at their first visit.

- Age (in months): mean at first visit ASD: 33, TD: 20.4
- The children had similar mean MLU at first visit: TD: 1.37, ASD: 1.24
- Mean Types at first visit (unique words used): ASD: 42, TD: 45
- Mean Tokens at first visit (amount of words used): ASD: 161, TD: 194

2. Describe linguistic development (in terms of MLU over time) in TD and ASD children (as a function of group).

Motivation for determining predictors

(Sigrid)

Fixed effects: As a measure of time, we have chosen to enter “age” and not “visit” in our model. Taken visit by visit, there is a skew in age since autistic children had a higher age at each visit than typically developing children. Thus, “visit” has a skewed representation of the children’s age. Furthermore, the time between each visit differs from visit to visit. Using ‘age’ as a predictor and not ‘visit’ accounts for this difference. However the limitation of entering age is that there is a difference in the maximal age for the two groups. TD have a max age of 45 months, whereas ASD have a max age of 62 months.

Random effects: We had random intercepts for each subject and random slopes for age to account for the subjects’ individual differences in both their skills at start and their different learning speed.

Eyeballing the data

Eyeballing the data gave the indication that there might be an interaction effect between age and diagnosis, and that the random effects are important especially for the ASD group where participants have very different developments (see figure 1).

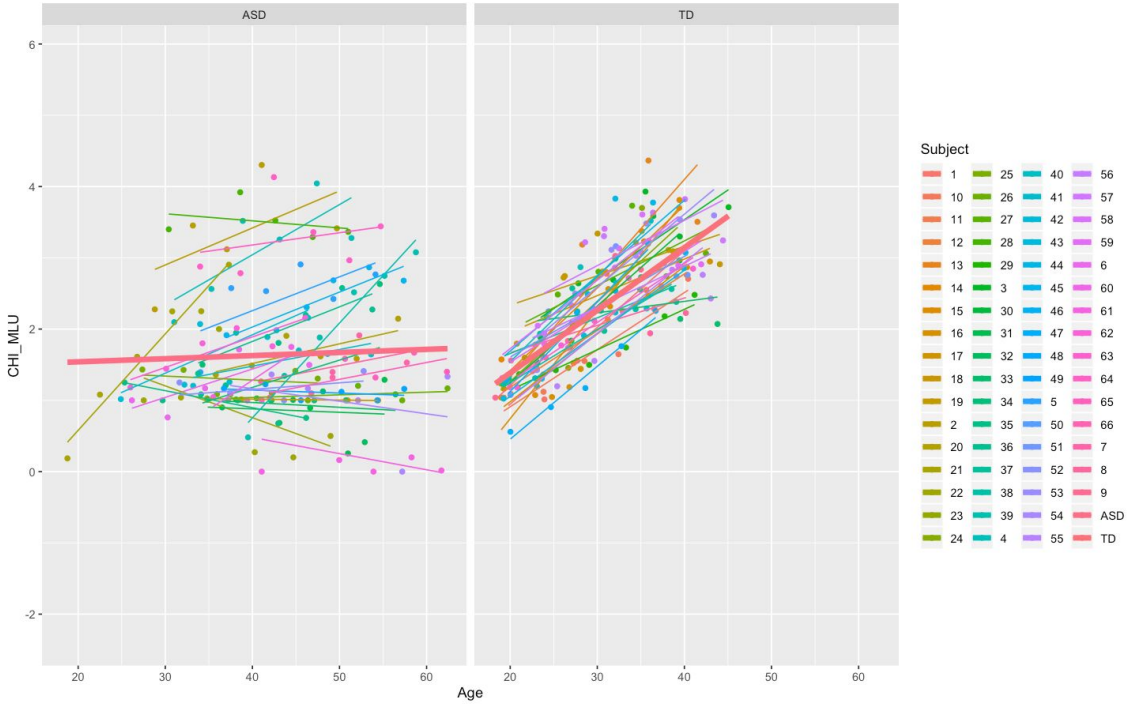

Figure 1: CHI_MLU plotted against age. Random slopes and intercept for each participant.

The model

(Anders) We have built a linear mixed effects growth model using the following syntax and package lme4:

*model = lmer(child_MLU ~ Diagnosis * Age + (Age|Subject) + I(Age^2))*

Results

(Anders) There was a significant interaction effect between diagnosis and age, $\beta = 0.030$, SE = 0.012, $t = 2.542$, $p < .01$.

The interaction effect makes the main effects of diagnosis and age, respectively, useless, since they can only be interpreted when taking the other into account.

Fixed effects alone accounted for roughly 38 % of the variance. Fixed effects and random effects together accounted for roughly 82 % of the variance.

An anova comparing our model with a model excluding the interaction Diagnosis:Age, showed that the interaction made our model significantly better (BIC = 681 and BIC = 612 respectively), thus confirming the importance of the interaction.

Description of results

(Gustav) Children who are typically developing have a significantly higher mean length of utterance as they develop compared to children who have an autistic spectrum disorder. For each month in their

development the mean length/utterance of typical children are 0.003 syllables longer than children with autism spectrum disorder. This is the interaction effect.

Plot of the models predictions

(Gustav) Figure 2 shows the actual values (blue) plotted against the predicted values (red). However these predictions are not sensible, since we didn't split the dataset into training and testing. The predicted values merely show that our model is good at explaining the data.

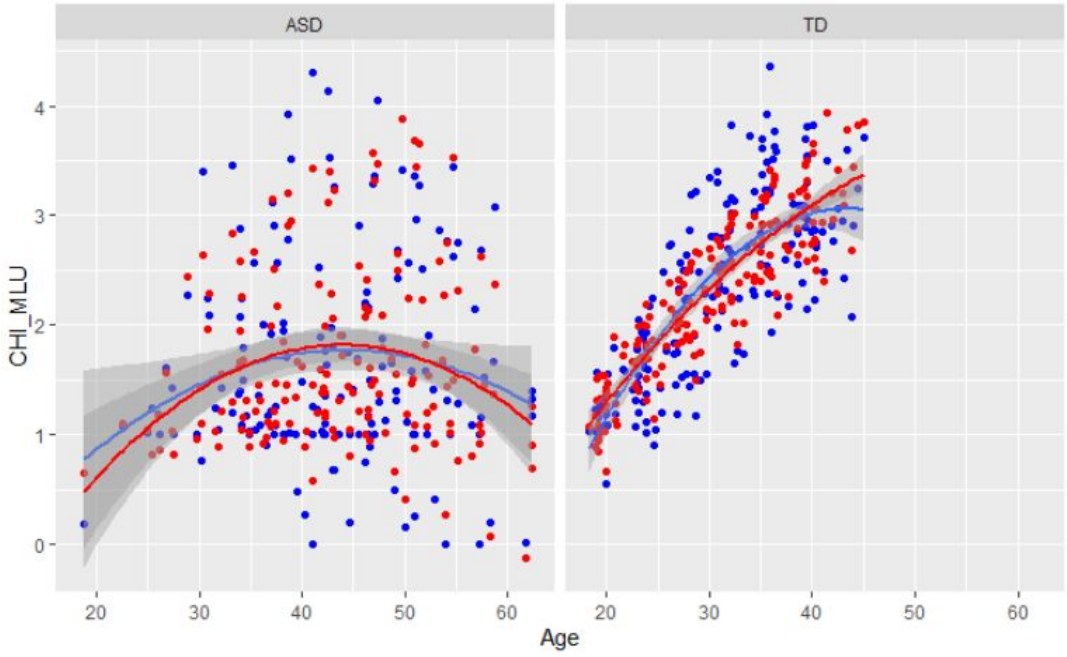

Figure 2: actual values of the mean length of utterance for children (blue) compared to fitted values (red).

3. Describe how parental use of language (in terms of MLU) changes over time. What do you think is going on?

Eyeballing the data

(Sigrid) Eyeballing the data showed that an interaction effect between age and diagnosis probably was not present. It also showed that random effects would not account for as much variance as it did with child mean length utterances.

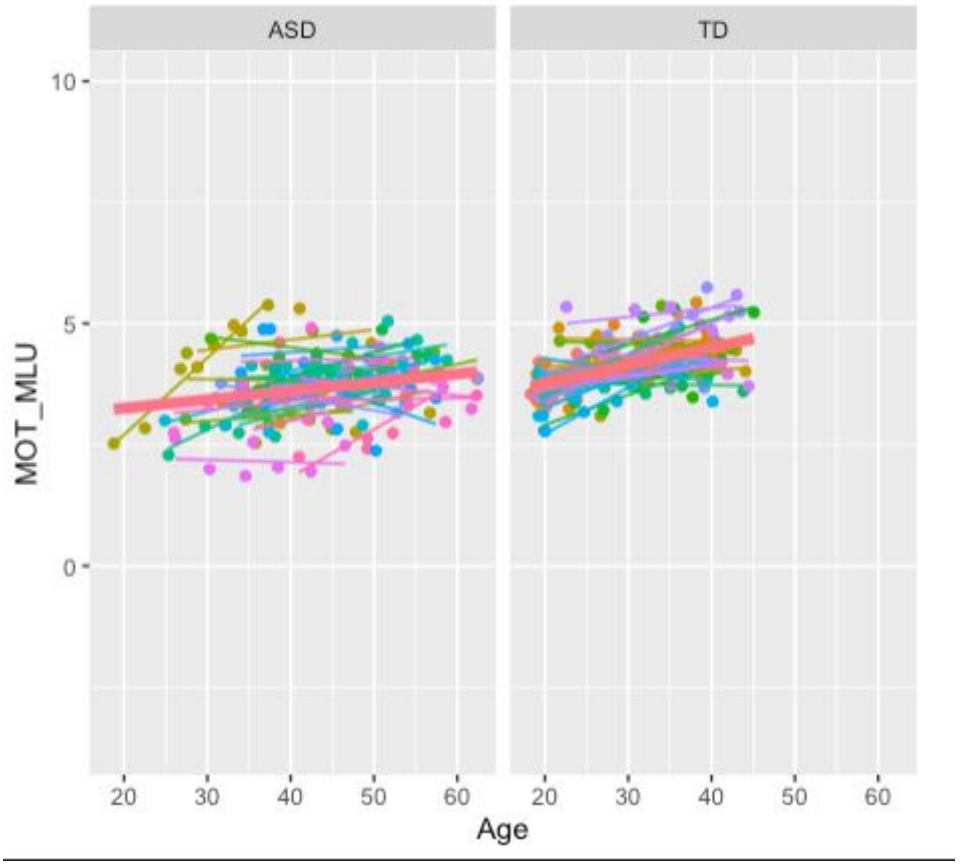

Figure 3: Mother MLU according to the diagnosis and age of the child

Results

(Sigrid) To assess the parental use of language (also in terms of MLU), a model was built using the following syntax:

model = lmer(Mother_MLU ~ Diagnosis + Age + (Age|SUBJ), df)

There was a significant effect of diagnosis; the mother of the child had a lower MLU if the child had the diagnosis of ASD, $\beta = 2.53$, SE = 0.18, $t(100.1) = 14.375$, $p < .0001$, compared to the MLU of mothers with typically developing children, $\beta = 3.314$, SE = 0.12, $t(61.73) = 6.31$, $p < .0001$. Moreover, there was a significant effect of age; the older the child got (in months) the larger MLU of the mother, $\beta = 0.0277$, SE = 0.004, $t(58.95) = 6.865$, $p < .0001$.

Fixed effects alone accounted for roughly 20 % of the variance. Fixed effects and random effects together accounted for roughly 68 % of the variance.

It seems that the mothers of typically developing children (TD) talk more to their children than mothers of children with ASD, and that the MLU of mothers increases as the age of the child increases. This makes sense since TD children have a larger MLU than ASD children.

The results are depicted in figure 4.

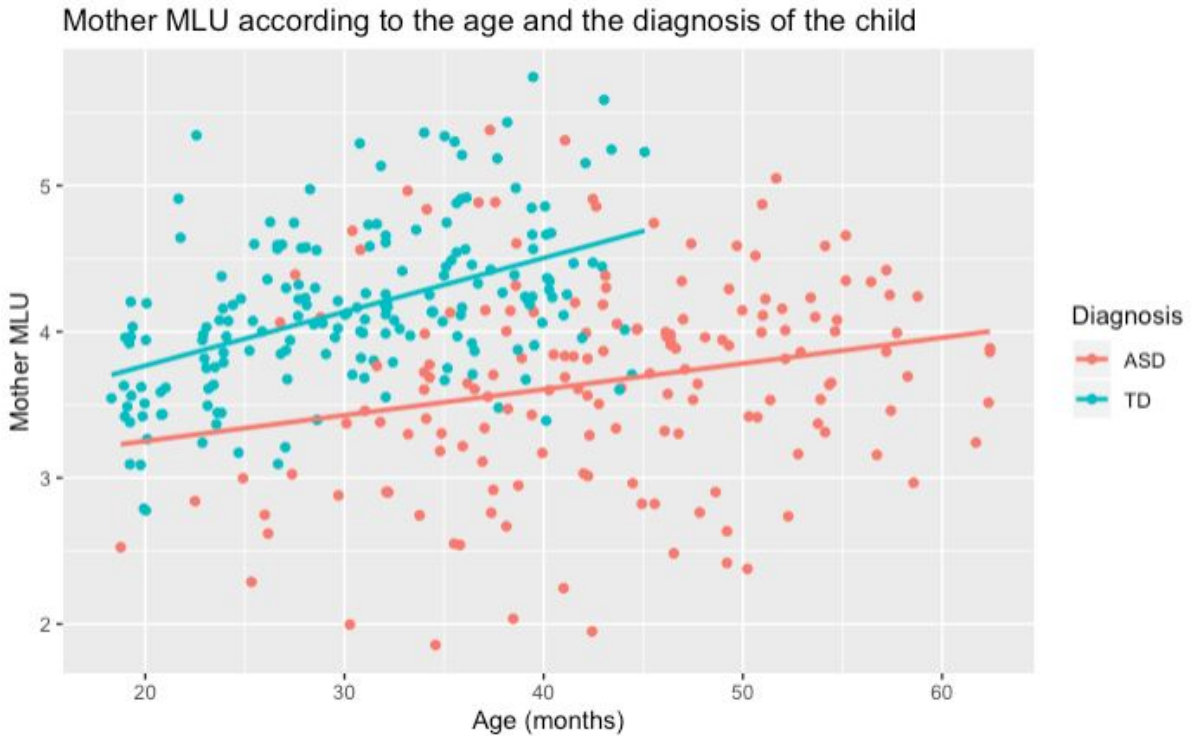

Figure 4: Mother MLU according to the age and diagnosis of the child

4. Include individual differences in your model of language development (in children). Identify the best model.

(Anders) We have conducted a post-hoc exploratory analysis to see the possible effect of other independent variables.
We found that 'ethnicity' was a significant predictor. However, this variable is not very well balanced in each level (e.g. there is only one asian subject), thus, we discard this variable as a valid predictor. 'Gender' was found to be non-significant.

The variables 'non-verbal-IQ', 'verbal-IQ' and 'Socialization' were all found to be significant. These results can be used to make new hypotheses in a follow up study. However, since we did not include them in our hypothesis we can't use them in the current analysis.

Assignment 2 part 2

Expmeth 3, Sigrid Snapfield, Gustav Helmet, Anders Seabridge Wheelman
Due 23/9 2020

The code can be found here:
https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_2_2

1. Discuss the differences in performance of your model in training and testing data

(Gustav) The model was built using the following syntax:
m8 <- CHI_MLU ~ Diagnosis * Age + (Age|Subject) + I(Age^2)

Testing the performance of the model on the data it was trained on vs test data

(Gustav)
Performance of model on the training data:
RMSE was calculated for fitted values to the model using the function fitted(). The training data was found to have an RMSE of 0.34 when compared to fitted values.

Performance of model on the test data:
RMSE was calculated for predicted values of the model using the function predict(). The training data was found to have an RMSE of 1.35 when compared to predicted values.

Thus, the model was performing better in the training data than in the test data. This makes sense since the model was created from the training data, meaning that the model is the best fit possible for the training data. Therefore it makes sense theoretically that the performance of the model on new data is not as good as its performance on its training data.

Cross validation results of model m8

(Sigrid) When cross validating the training data using 10 folds, we get a training RMSE value of 0.3524 and a test RMSE value of 0.6765. The RMSE scores of the model after it has been

cross-validated are approximately half the size of the RMSE scores for the model that has not been cross validated several times. These results suggest that cross validation gives less prediction error and is thereby a better method for testing models.

2. Which individual differences should be included in a model that maximizes your ability to explain/predict new data?

(Anders) In addition to the model mentioned above (m8), we created 7 other models:

```
m1 <-  CHI_MLU ~ Diagnosis + Age + (1|Subject)

m2 <-  CHI_MLU ~ Diagnosis + Age + (Age|Subject)

m3 <- CHI_MLU ~ Diagnosis + Age + (1|Subject)+ I(Age^2)

m4 <- CHI_MLU ~ Diagnosis + Age + (Age|Subject)+ I(Age^2)

m5 <- CHI_MLU ~ Diagnosis * Age + (1|Subject)

m6 <- CHI_MLU ~ Diagnosis * Age + (Age|Subject)

m7 <- CHI_MLU ~ Diagnosis * Age + (1|Subject)+I(Age^2)
```

We made a 10-fold cross-validation to compare the performance of all the models (see table 1). We found that the rmse-value for test-data was best (i.e. lowest) for m8, the fully complex model.

Thus, the most complex model is the best fit to new data which it has not been trained on. Normally, we might think that the most complex model would overfit and not generalize well to new data. However, here we find that it is indeed the most complex model which generalizes the best.

Type	Train	Test
rmse_m1[11]	0.4674175	0.7781820
rmse_m2[11]	0.3505257	0.8391998
rmse_m3[11]	0.3943179	0.6950842
rmse_m4[11]	0.3425541	0.6954798
rmse_m5[11]	0.4100964	0.7132800
rmse_m6[11]	0.3613275	0.6956282
rmse_m7[11]	0.3918088	0.6937753
rmse_m8[11]	0.3509706	0.6972461

Table 1: mean rmse-values for cross-validation of model m1-m8 with 10 folds.

Bonus Question 1: What is the effect of changing the number of folds? Can you plot RMSE as a function of number of folds?

(Gustav) In general, the more folds, the lower the value of RMSE. Thus, the more folds the better (given that the data set is big enough). However, when increasing the number of folds on the training data set, the RMSE value does not seem to decrease. It might be due to outliers (see figure 1).

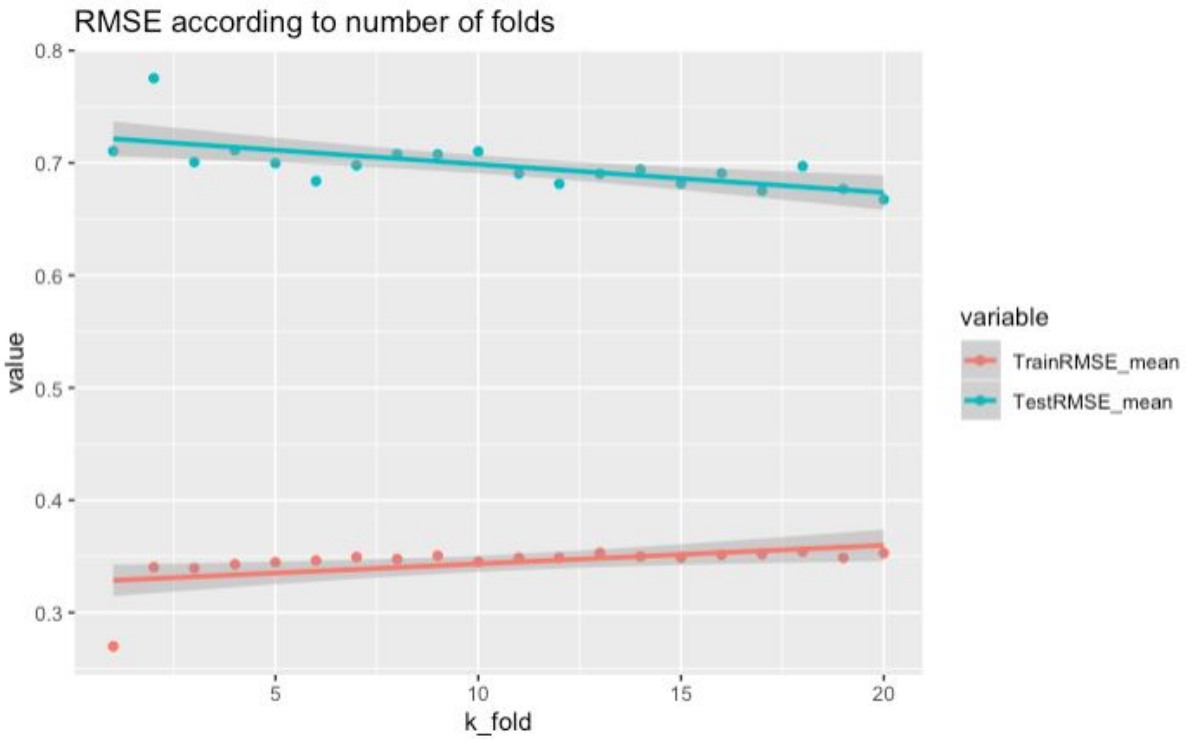

Figure 1: mean rmse-values for training and test data as the number of folds increases.

Bonus Question 2: compare the cross-validated predictive error against the actual predictive error on the test data

(Sigrid) The mean predictive error for the test data was 0.98.
We calculated this value by subtracting the predicted values of m8 by the test data and taking the mean of these values.

The mean predictive error for the cross validated model was 0.79. We calculated the value by extracting the average predictive error for each fold and then getting the mean of the average predictive error for each fold

Thus, the cross-validated predictive error is lower than the actual predictive error of the test data. This confirms that cross validation is a good method for lowering the predictive error.

3. Predict a new kid's performance (let's call him Bernie) and discuss it against expected performance of the two groups

(Anders) The absolute difference between the MLU of Bernie (subject 2) and the MLU of typically developing children at each visit can be found in the column 'Absolute_dif' in table 2.

Subject <dbl>	Diagnosis <chr>	Visit <dbl>	CHI_MLU <dbl>	mean_td <dbl>	mean_asd <dbl>	Age <dbl>	Absolute_dif <dbl>
2	ASD	1	1.984456	1.310174	1.304476	27.80	0.6742815
2	ASD	2	2.544444	1.778062	1.443683	33.33	0.7663823
2	ASD	3	3.353191	2.229037	1.799177	36.23	1.1241544
2	ASD	4	3.183099	2.734681	1.815197	40.03	0.4484175
2	ASD	5	3.173252	2.969893	1.598512	45.47	0.2033593
2	ASD	6	3.448413	2.927875	1.940044	49.30	0.5205373

Table 2: The data of Bernie (subject 2) including the absolute difference of child MLU between Bernie and TD children.

It is clear that the MLU of Bernie is better than both the averaged MLU of typically developing children and that of children with autism spectrum disorder (ASD) (figure 2).

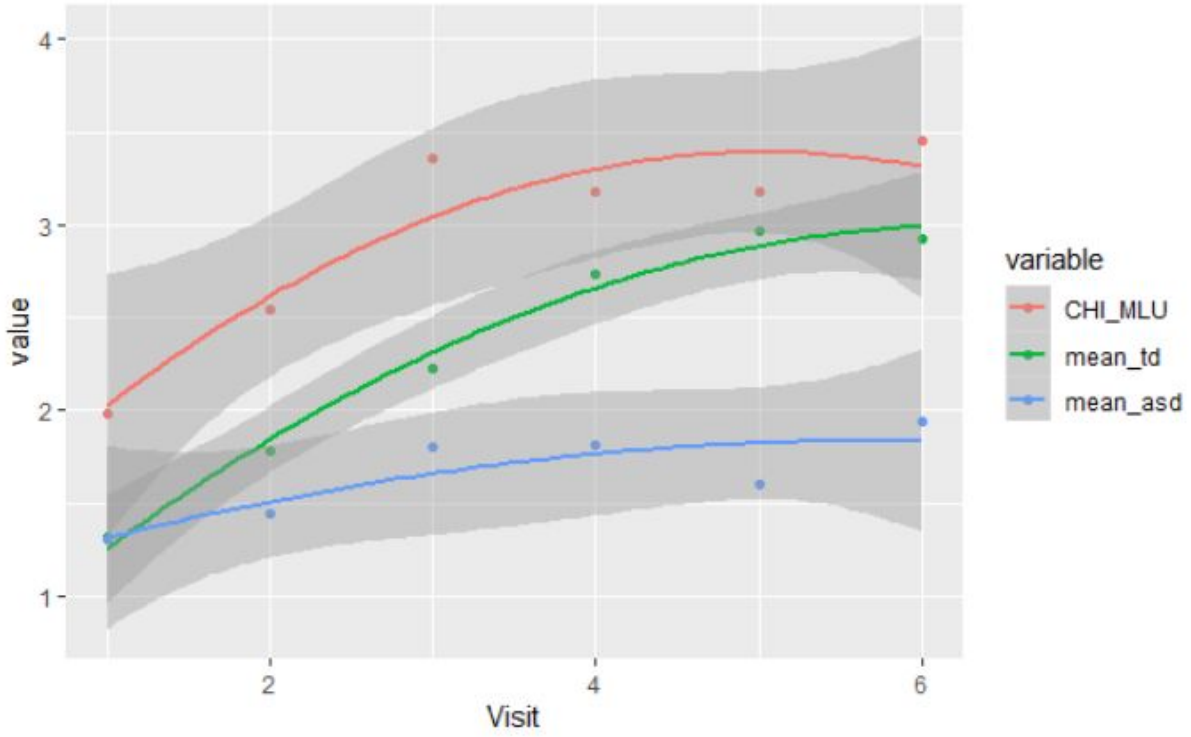

Figure 2: mean CHI_MLU of Bernie is better across visits than mean MLU of typically developing children and autistic children.

(Gustav) When using our model to predict the MLU of Bernie at visit 6, we find that Bernie is doing better than expected. The predicted MLU of Bernie at visit 6 is 1.5988 phonemes, while the actual MLU of Bernie at visit 6 is 3.4484. Thus, Bernie has a MLU which is 1.8496 higher than the predicted MLU of ASD children at visit 6. This is a good example of why it is important to remember that our model is just a model, and there will always be individual differences and thereby subjects who stand out.

Assignment 2 part 3

Expmeth 3, Sigrid Snapfield, Gustav Helmet, Anders Seabridge Wheelman
Due 30/9 2020

The code can be found here:
https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_2_3
The file is called 'FOR_GRADING_A2_P3'

In this part of the assignment, we try to figure out how a new study should be planned (i.e. how many participants are needed) in order to have enough power to replicate the findings (ensuring our sample size is adequate, our alpha at 0.05 and our beta at 0.8):

Introduction

(Sigrid) We will be working with two datasets: dataset A, which entails a full dataset consisting of 346 rows (66 participants), and dataset B, which functions as a "pilot" dataset consisting of 36 rows (6 participants). In the preprocessing of the data, 26 rows were excluded in dataset A by criteria of missing values in the variables "CHI_MLU" (child MLU) and "age".

All power analyses were conducted with the following model using the package lme4 (Bates, Maechler, Bolker & Walker, 2015).

model1 = child_MLU ~ Diagnosis * Visit + (Visit|Subject)

From the previous parts of this assignment we know that a significant effect of the interaction between Diagnosis and Visit can be found. Thus, we will only examine the interaction effect and thereby disregard the main effects when conducting the following power analyses.

All power analyses were conducted by running 50 simulations using the package simr (Green & MacLeod, 2016) to test the sufficient number of participants needed to get a power of at least 80% at an alpha level of 0.05.

1. if we trust the estimates of the current study. Report the power analysis and comment on what you can (or cannot) use its estimates for.

(Anders) A power analysis on dataset A using model1 showed that the interaction Diagnosis:Visit had a power of 100 %, CI 95 % [93, 100]. In figure 1 it can be seen that 25 participants are needed to obtain sufficient power of 80%. The current study has 66 participants. We can thereby conclude that the current study was overpowered.

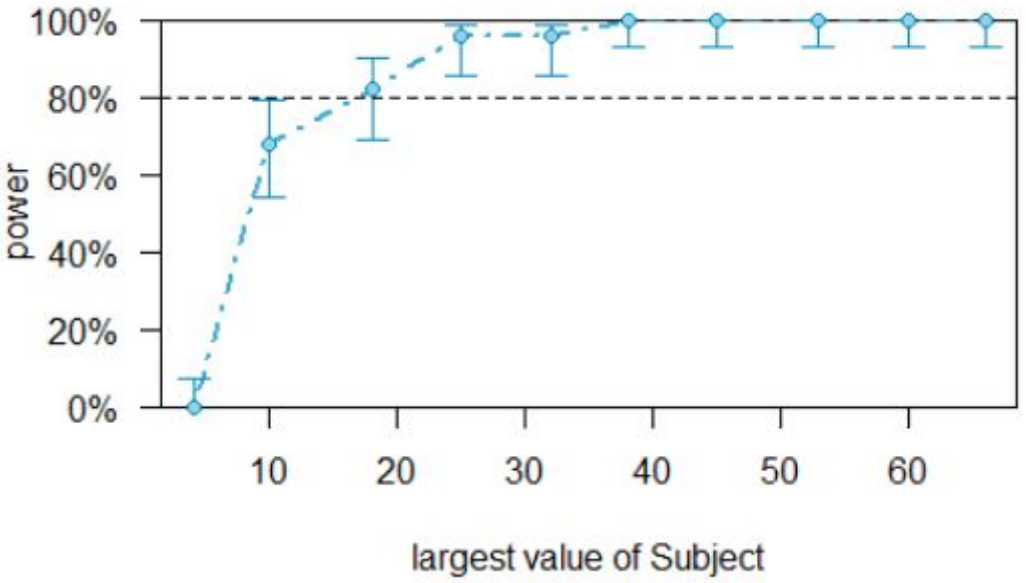

Figure 1: Power plot on dataset A. The effect size of Diagnosis:Visit is 0.25. 25 participants are needed to obtain a power of >80%.

(Gustav) The estimate of Diagnosis:Visit is 0.25. Thus, typically developing children will have a mean length of utterance that is 1.5 syllables longer than children with an autism spectrum disorder at visit 6. We argue that this estimate is too high, and would not necessarily be replicated in a follow-up study. Therefore, we lower the estimate for a more conservative power analysis.

2. if we are skeptical of the current study. Report the power analysis and comment on what you can (or cannot) use its estimates for.

Full study

(Sigrid) In order to run a more conservative power analysis the effect size can be set to a lower value. However, the effect size should still be theoretical justifiable. We have decided that a difference of minimum 1 syllable per utterance from visit 1 to visit 6 is sufficient for a meaningful effect size. Thus, the lowest justifiable effect size would be 1 syllable divided by 6 visits, which equals 0.167.

A power analysis was conducted on model 1 by simulating dataset A at an effect size of 0.167 (see the section 'Introduction' for details) . Results showed that a total sample of 53 participants was required to achieve a power of 94% with CI 95% [83.45, 98.75] (figure 2).

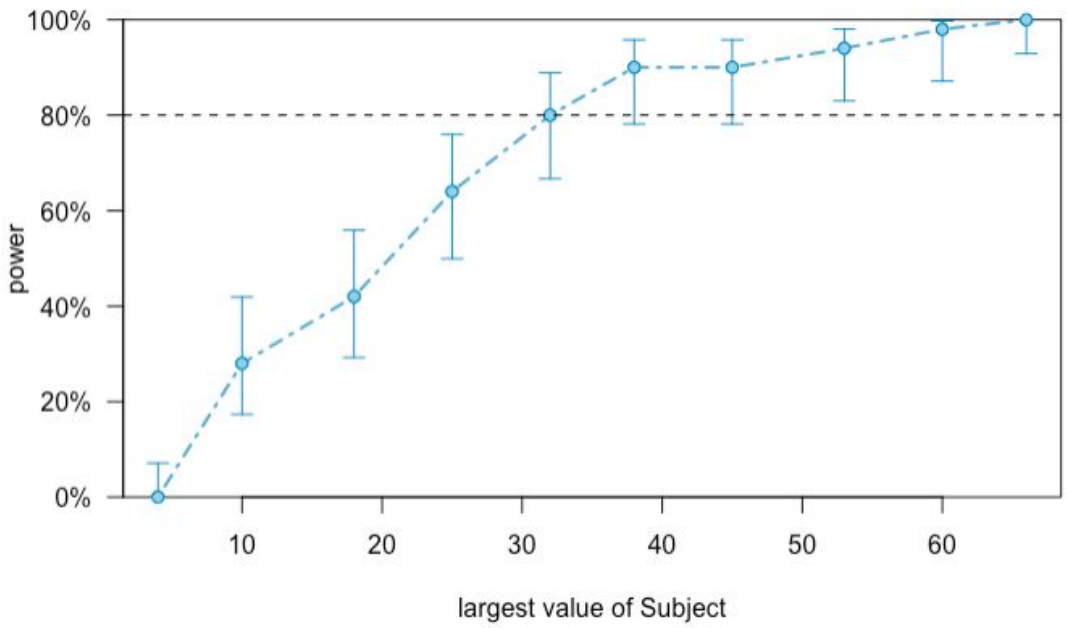

Figure 2: Power curve for an effect size of 0.167 simulating dataset A on model 1. 53 participants are sufficient to get a power of >80%.

Pilot study

(Anders) Since we had access to another dataset (dataset B) we decided to analyse it as it was a pilot study with 6 participants. Note that this is not a legit pilot-study, but merely an exploratory exercise to investigate the validity of a power analysis on a pilot study.

First we conducted a power analysis on dataset B. We found that with 6 participants and an effect size of 0.167 the pilot study had a power of 10 % with CI 95% [3.33, 21.81]. We then conducted a power analysis where the number of participants had been extended, to investigate the sufficient number of participants needed to obtain a power of >80%. We found that a power of 92% with CI 95% [80.77,97.78] were reached with 48 participants (figure 3). This estimate was a good approximation of the estimate of 53 participants in our analysis using dataset A. However, it is important to take into account that it is assumed that the pilot study data is representative of the population, which might be violated since dataset B only had 6 participants, that is n = 3 in each group (ASD, TD).
All in all this experiment supports the notion that extending the number of subjects from a pilot study gives a good approximation of the number of participants needed to obtain a power of >80%.

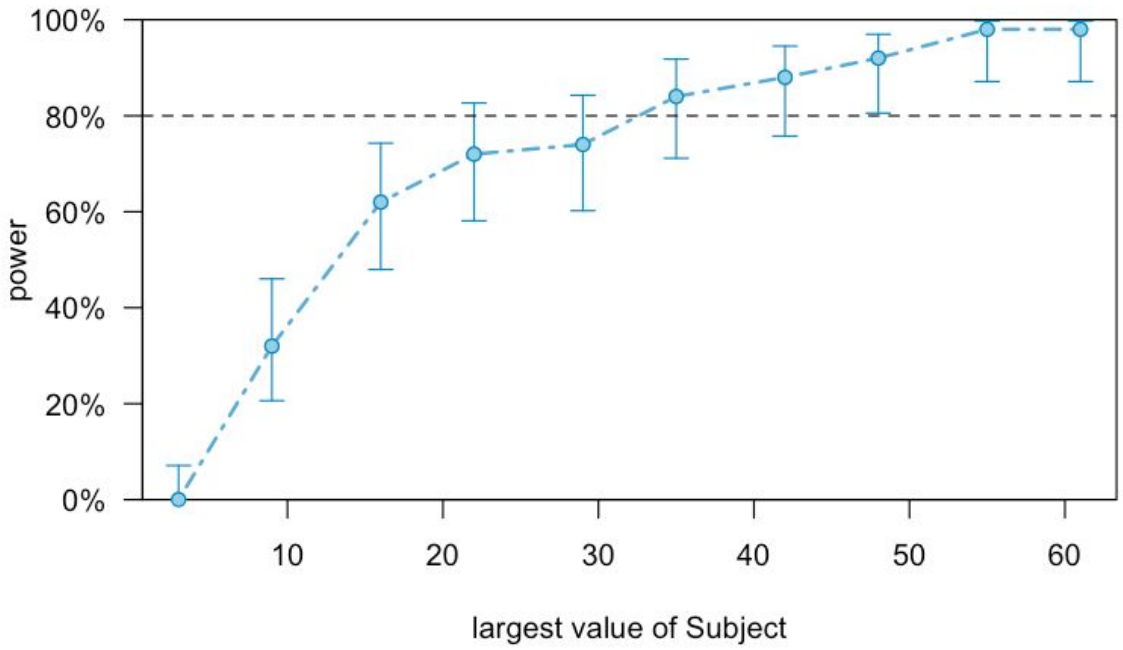

Figure 3: Power curve for an effect size of 0.167 simulating dataset B on model 1 as a pilot study. 48 participants are sufficient to get a power of >80%.

3. if we only have access to 30 participants. Identify the power for each relevant effect and discuss whether it's worth to run the study and why

(Gustav) We assessed the power for the interaction effect between Diagnosis and Visit with n = 30 using the function extend(n=30). We found that with 30 participants and an effect size of 0.167, the study had a power of 86 % with CI 95% [73.26 , 94.18]. The obtained power of 30 participants is insufficient because of the large confidence interval that drops below 80%.
It is therefore not worth it to run the study with only 30 participants.

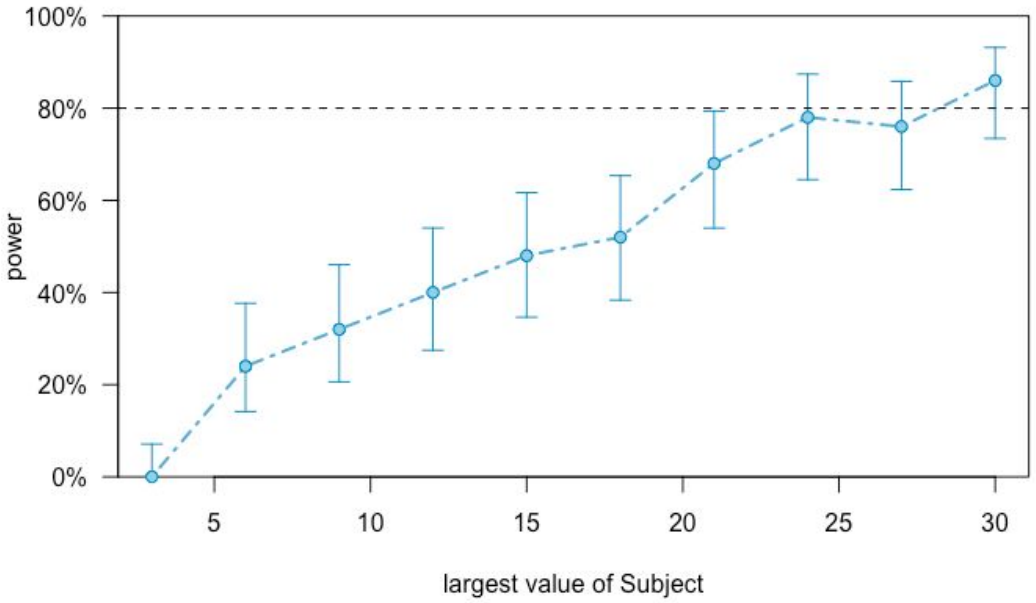

Figure 4: Power curve for the dataset with 30 participants.

Extending visit

(Sigrid) We cannot recruit more participants to gain more power, but we can extend the number of visits to gain more power. Figure 5 shows the result of extending to 10 visits and it shows that 7 visits with 30 participants would give a power of 92.00% with CI = [80.77, 97.78].

However, we know from the previous parts of the assignment that the effect of MLU is best described by a growth curve model rather than a linear model. This means that the effect will stagnate along visits. Model 1 uses the linear method, why extending visit will give an inaccurate estimate of power for the true effect.

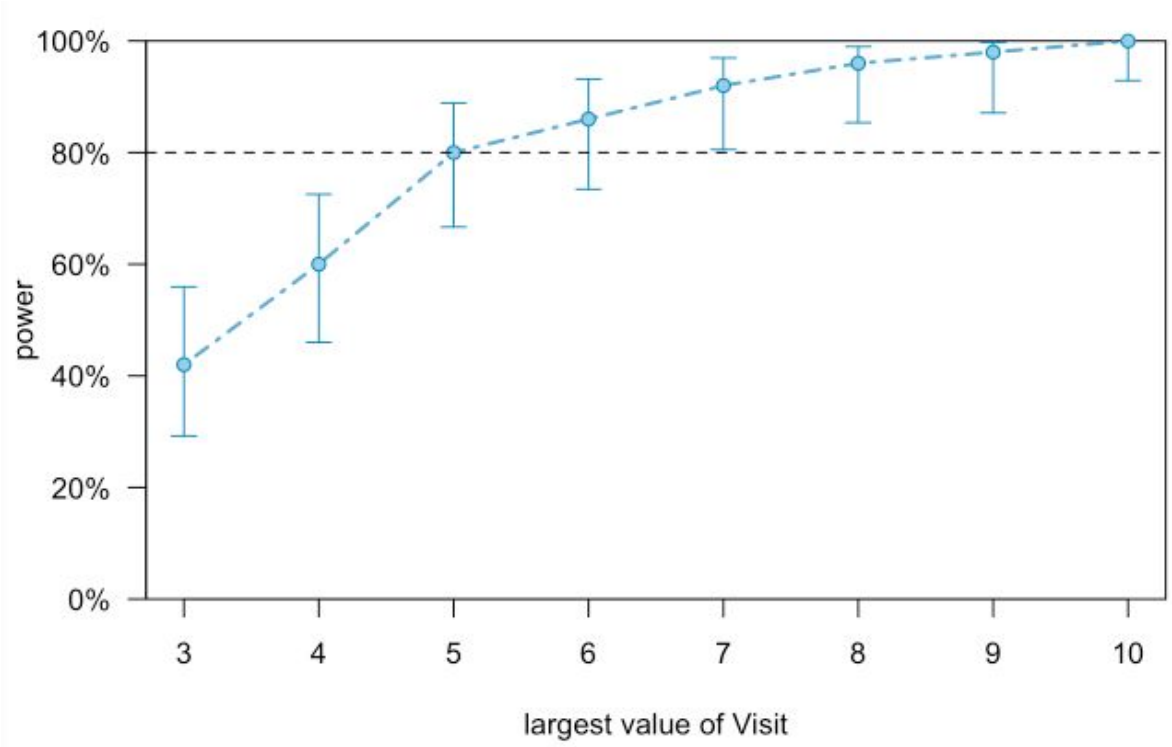

Figure 5: Power curve for an effect size of 0.167 simulating dataset A on model 1 with only 30 participants. Extending visit to 7 is sufficient to get a power of >80% with only 30 participants.

References

- Bates, Maechler, Bolker & Walker (2015). Fitting Linear Mixed-Effects Models Using lme4. Journal of Statistical Software, 67(1), 1-48. doi:10.18637/jss.v067.i01.

- Green & MacLeod (2016). simr: an R package for power analysis of generalised linear mixed models by simulation. Methods in Ecology and Evolution, 7(4), 493-498. doi: 10.1111/2041-210X.12504 (URL: https://doi.org/10.1111/2041-210X.12504), <URL: https://CRAN.R-project.org/package=simr>.

Assignment A3 part 1 - due 7/10

Study group: ANDERS WHEELMAN, GUSTAV HELMET AND SIGRID SNAPFIELD.

Code link: https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_3_1

Methods

Participants (ANDERS)

We have gathered data from four Danish studies (study 1-4) and two Mandarin studies (study 5-6). We have excluded the Japanese study (study 7).
The data set consisted of 3330 observations (where n women = 1452, and n men = 1870, NA = 8). The mean age for women across groups is 27.6 years and the mean age for men across groups is 27.9 years.

The two groups (control and diagnosis) are fairly balanced (table 1). However, we find a salient difference in Total IQ mean between the two groups, that might cause bias in the analysis.

	Control group (n = 1801)	Schizophrenia group (n = 1529)
n women	738	714
n men	1055	815
mean age (years)	28.3	27.1
SANS mean, min, max	0.39 [0, 7]	7.7 [0, 20]
SAPS mean, min, max	0.07 [0, 3]	9.95 [0, 20]
Total IQ mean	102.9	90.47
Education mean	14.9	12.78

Table 1: Summary of important variables within the groups, control and diagnosis.

The studies have different numbers of participants. Study 1-4 and study 6 have fairly balanced groups but study 5 is very skewed (table 2).

	n in total	n control	n schizophrenic
Study 1	683	348	335
Study 2	363	184	179
Study 3	387	236	151

Study 4	471	233	238
Study 5	201	201	0
Study 6	1225	599	626

Table 2: Number of participants in each study.

The only systematic difference we have found is the skewed distribution of study 5, since it only contains subjects in the control group.
However, there might still be unsystematic differences across the studies.

Statistical procedures

Pre-processing (SIGRID)

The full dataset consists of 7 studies conducted in three different languages. We have filtered out study 7 as it caused problems in the analysis. Thus, we have excluded Japanese participants and will only be looking at Danish and Chinese subjects.

Moreover, we have filtered out rows where “Diagnosis”, “duration”, “phonation time” and “number of pauses” have missing values. This reduces the dataset from 3401 observations to 3300 observations.

Variables (GUSTAV)

The objective of this analysis is to replicate and compare our results with those of the meta-analysis of Parola, A., Simonsen, A., Bliksted, V., & Fusaroli, R. (2019). In order to create the appropriate models for investigating whether it was possible to replicate the results in the meta analysis, we had to recreate their significant variables: pitch variability, proportion of spoken time, speech rate and pause duration. The variables were calculated as follows:

Pitch variability

- The pitch variability variable was calculated by taking the highest pitch frequency (Hz) minus the lowest pitch frequency (Hz) for each recording. This gives the span of pitches participants used.

Proportion of spoken time

- The proportion of spoken time variable was calculated by taking the time spoken in each recording divided by the total duration of the recording.

Speech rate

- The speech rate variable was calculated by dividing the total amount of syllabus in each recording with the total duration of the recording.

Pause duration

- The pause duration variable was calculated by dividing the duration where no speaking was recorded (i.e. total duration minus time spoken) with the total amount of pauses for each recording.

All variables were scaled in order to make the output of the analysis comparable to the results of the meta analysis using the scale()-function in R (R Core Team, 2019). The scale function subtracts the mean from the observation and divides by standard deviation.

Making Models (ANDERS)

We have made 4 generalized linear mixed effect models to investigate whether the pre-mentioned 4 variables predict diagnosis. We included both subject and study as random intercepts since we expected unsystematic variance between each subject and each study. We built them using the following syntax:

Diagnosis ~ variable + (1|Subject) + (1| Study), family = “binomial”

Prior to making these 4 models, we investigated whether there was a difference in the respective variable in the two languages. We built the following linear mixed effect model for each of the 4 variables, thus making a total of 4 models:

variable ~ Language + (1|Subject) + (1| Study)

If there was a significant difference in the respective variable we separated the analysis according to language. However, if there was not a significant difference, we pooled both languages together in the same analysis.

Converting estimates into probability (SIGRID)

We built a function to convert estimates from the generalized linear mixed effect models into probabilities. The function was built by the following formula:

F(1) <- probability = $e^{logit} / (1 + e^{logit})$

Where “logit” denotes the value of the estimate (e.g. $b_0 + 3 * b_1$).
As we will explain in our results-section, this function caused some problems.

Investigating Hedge’s g (GUSTAV)

In order to compare the effect sizes of our model to the results of the meta-analysis, we carried out a hedges g calculation. We calculated hedges g for each of the 4 variables using the package “effsize” (Torchiano M, 2020) and the function cohen.d(), specifying a hedges.correction = T.

Results

Pitch variability (ANDERS)

Language was not found to be a significant predictor or pitch variability, t(3321) = 1.86, p = .13. Thus, we have not separated the data set into Danish and Chinese, respectively, in the following analysis.

Pitch variability was found to significantly predict diagnosis, z(3329) = -175.6, p < .001.

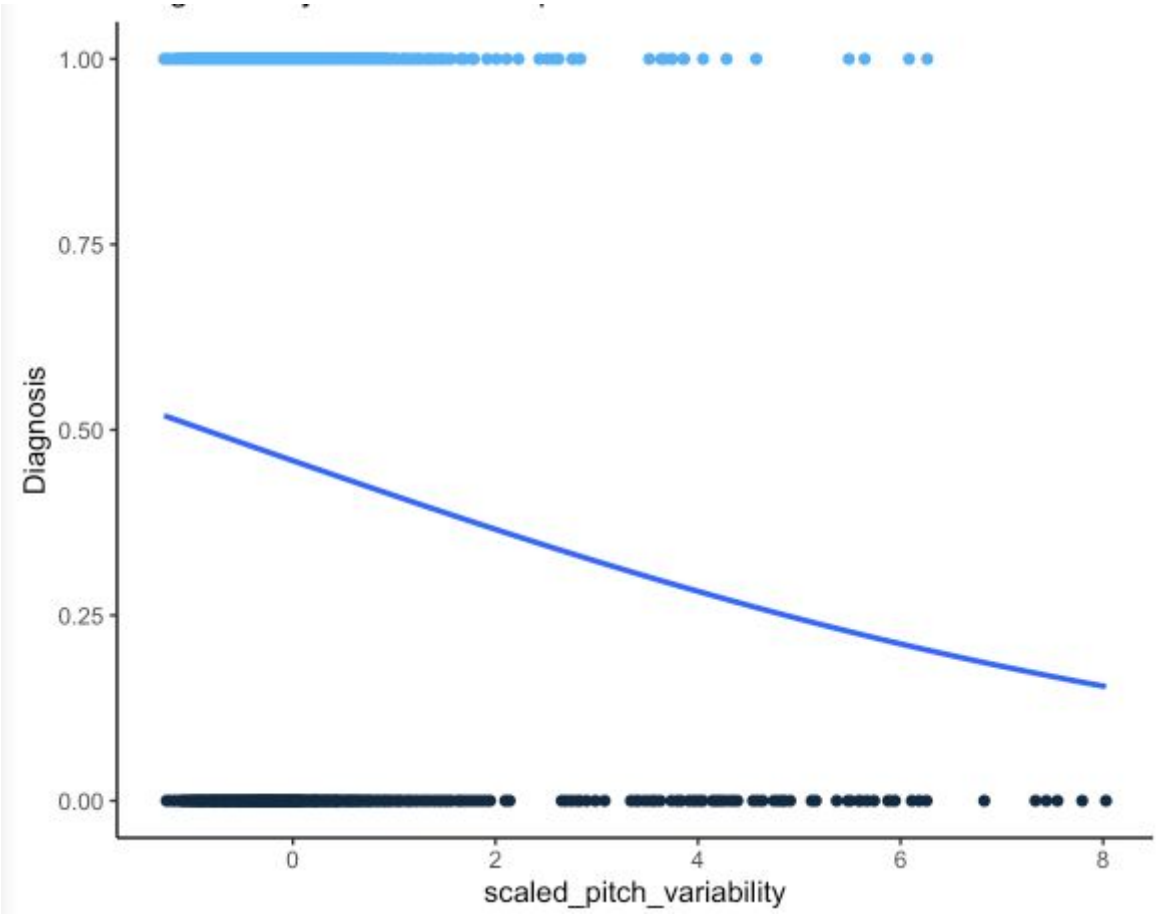

Figure 1: Diagnosis predicted by scaled pitch variability.

To convert model estimates into probabilities, we entered the intercept as the logit-value in the function mentioned above F(1). We found that the probability of being schizophrenic at a scaled pitch variability of 0 was 99.9%.

We found that scaled pitch variability had the following standardized effect size: Hedges' g: 0.18, 95% CI[0.11, 0.25].

Proportion of spoken time (SIGRID)

Language was found to be a marginally significant predictor for speech rate, t(3.9) = - 2.63, p = .0593. We have not separated the data set into Danish and Chinese, respectively, in the following analysis.

Proportion of spoken time was not found to significantly predict diagnosis, z(3329) = - 0.37, p = .72.

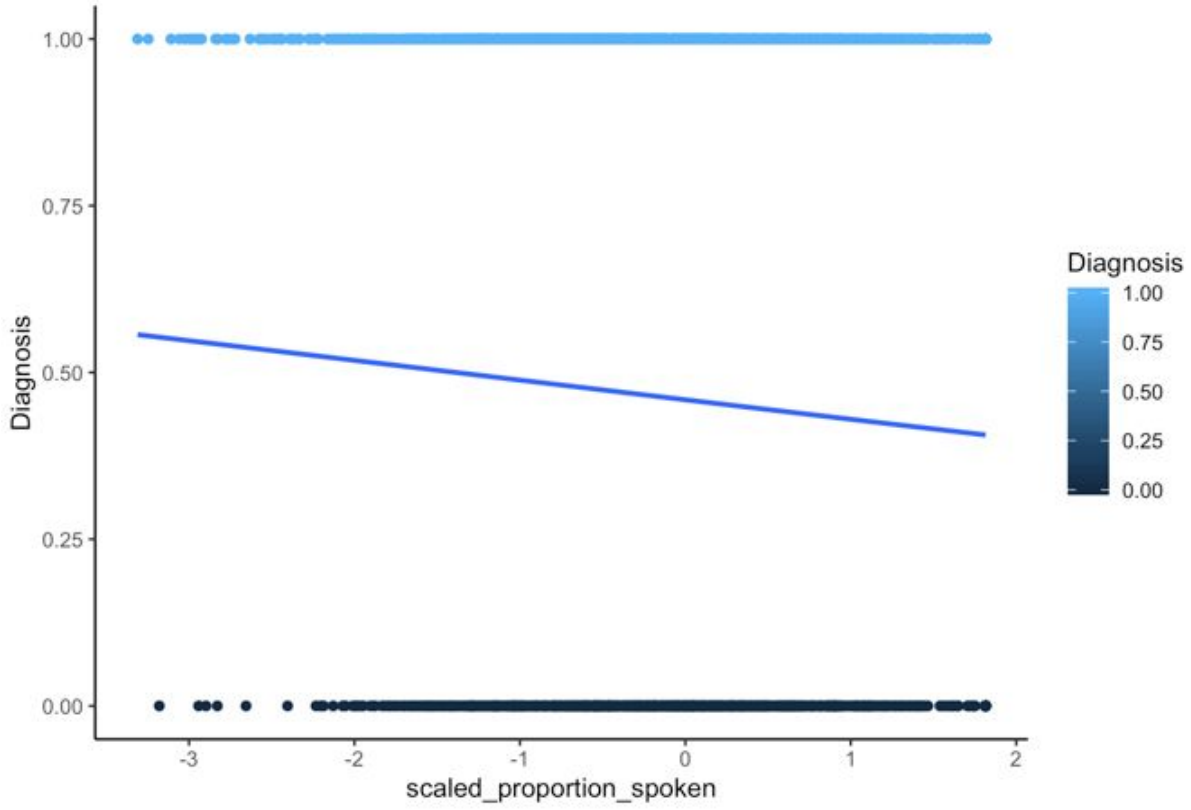

Figure 2: Diagnosis predicted by scaled proportion of spoken time.

To convert model estimates into probabilities, we entered the intercept as the logit-value in the function mentioned above F(1). We found that the probability of being schizophrenic at a scaled speech rate of 0 was 99.9%.

We found that scaled pitch variability had the following standardized effect size:
Hedges' g: 0.12, 95% CI[0.05, 0.19].

Speech rate (GUSTAV)

Language was not found to be a significant predictor for speech rate, t(4.2) = -1.86, p = .13. Thus, we have not separated the data set into Danish and Chinese, respectively, in the following analysis.

Speech rate was not found to significantly predict diagnosis, z(3329) = - 0.39, p = .69.

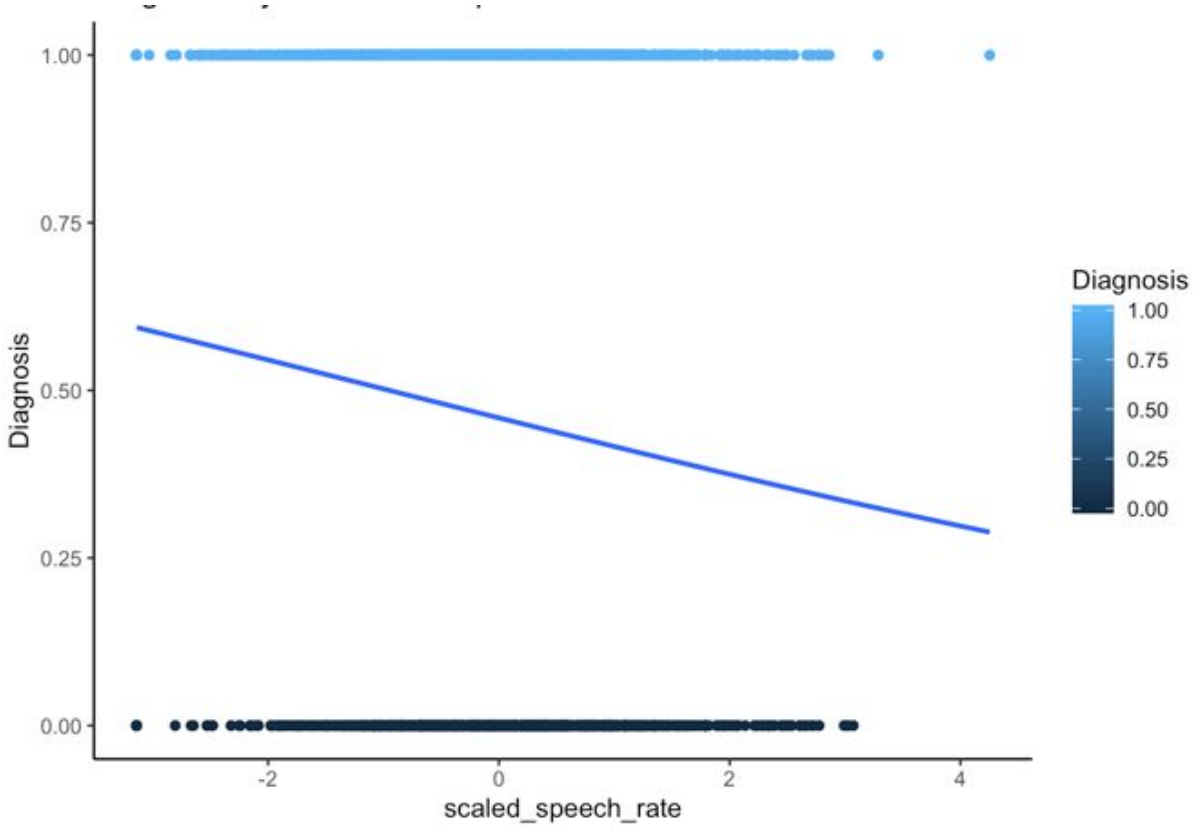

Figure 3: Diagnosis predicted by scaled speech rate.

To convert model estimates into probabilities, we entered the intercept as the logit-value in the function mentioned above F(1). We found that the probability of being schizophrenic at a scaled speech rate of 0 was 99.9%.

We found that scaled pitch variability had the following standardized effect size:
Hedges' g: 0.17, 95% CI[0.10, 0.24].

Pause duration (ANDERS)

Language was found to be a significant predictor or pause duration, $t(3321) = 2.98$, $p = 0.046$.. Thus, we have separated the data set into Danish and Chinese.
We ran the logistic model on the Danish subjects, and found that there were no significant differences in pause duration according to diagnosis.
However, when running the model on chinese subjects, we found a significant difference in pitch variability between the two diagnoses. Thus, we continued our analysis only on chinese subjects.

Pause duration was found to significantly predict diagnosis in chinese subjects, $z(1425) = -53.24$, $p < .001$.

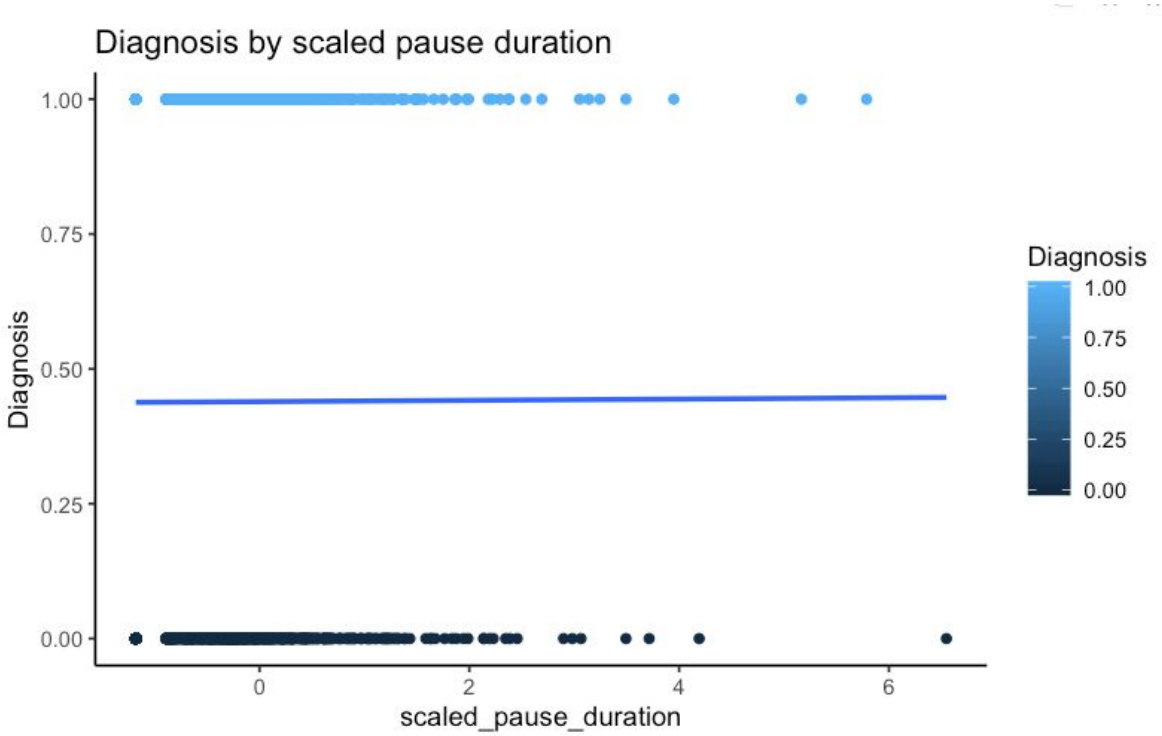

Figure 4: Diagnosis predicted by scaled pause duration for Chinese subjects.

To convert model estimates into probabilities, we entered the intercept as the logit-value in the function mentioned above F(1). We found that the probability of being schizophrenic at a scaled pause duration of 0 was 99.9%.

We found that scaled pitch variability had the following standardized effect size: Hedges' g: -0.004, 95% CI[-0.11, 0.10].

Discussion

The features of a schizophrenic voice (SIGRID)

Pitch variability was found to significantly predict the diagnosis of subjects. Thus, the lower the pitch variability the higher the chance of being schizophrenic and vice versa, the higher the pitch variability the lower the chance of being schizophrenic. Thus, the results indicate that low pitch variability, i.e. a monotone voice is a feature of a schizophrenic voice.

Moreover, pause duration was found to significantly predict diagnosis for Chinese subjects. The higher the pause duration, the greater probability of being schizophrenic. This might indicate a cultural difference, since it was not found to be significant for the Danish subjects.

However, we suspect that the function used to convert model estimates to probabilities is not functioning correctly. The probability of being schizophrenic at the intercept for scaled pitch-variability is 99.9%. Identically, the probability of being schizophrenic at a scaled pitch-variability of 10 is also 99.9%. Thus, we have a hard time concluding anything about

the direction of the effect of pitch variability and consequently the features of a schizophrenic voice (e.g. whether it has a lower or higher pitch) when converting to probabilities.

Thus, due to both the probabilities acting weird and the effect sizes being very small, it is difficult to conclude anything.

Relating results to meta-analytic findings (GUSTAV)

We found significant results in the variables of pitch variability and pause duration, though the latter only applied to the Chinese subjects (see table 3). The found effects were, however, much smaller than those reported in the meta-analysis paper of Parola et al. (2019).

All directions of our Hedge's g were opposite of those from the meta-analysis. This might be due to the meta-analysis having a different leveling of their outcome variable (Diagnosis) than we had in our analysis. However, we were not able to obtain the leveling of the Diagnosis variable in the meta-analysis and cannot say whether the difference is due to different leveling of diagnosis.

In conclusion, our analysis did not successfully replicate the findings of the significant variables found in the meta-analysis.

	Variable	Hedge's g	Significance level
Our study	Pitch variability	0.18, 95% CI [0.11, 0.25]	**p < .001**
Meta-analysis		- 0.55, 95% CI [-1.06, 0.09]	**p = .005**
Our study	Proportion of spoken time	0.12, 95% CI [0.05, 0.19]	p = .72
Meta-analysis		-1.26, 95% CI [-2.26, 0.25]	**p = .001**
Our study	Speech rate	0.17, 95% CI [0.10, 0.24]	p = .69
Meta-analysis		-0.75, 95% CI [-1.51, 0.04]	**p = .015**
Our study (only chinese subjects)	Pause duration	-0.004, 95% CI [-0.11, 0.10]	**p < .001**
Meta-analysis		1.89, 95% CI [0.72, 3.21]	**p < .001**

Table 3: The results of our analysis compared to the meta analysis. Significant p-values are in bold.

References

- Parola, A., Simonsen, A., Bliksted, V., & Fusaroli, R. (2019). Voice patterns in schizophrenia

- R Core Team (2019). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL https://www.R-project.org/.

- Torchiano M (2020). *effsize: Efficient Effect Size Computation*. doi: 10.5281/zenodo.1480624, R package version 0.8.1, https://CRAN.R-project.org/package=effsize.

Assignment A3 part 2 - due 4/11

Diagnosing Schizophrenia from Voice features

Study group: ANDERS WHEELMAN, GUSTAV HELMET AND SIGRID SNAPFIELD.
Code link: https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_3_2

Methods

Participants

See assignment A3 P1 for a description of participants. (Anders)

Dataset (ANDERS)

The data included two languages (Danish and Chinese) and spanned across 6 studies, where study 1-4 were conducted in Danish, and study 5-6 were conducted in Chinese.
Before analyzing the data, the dataset was split into a training set (80%) and a test set (20%) using the partition function from the groupdata2 package (Olsen, 2020) in R (R Core Team, 2019). The function ensures an equal proportion of the two diagnoses in each data set. (Gustav)

The model and feature preprocessing (SIGRID)

The model used in the analysis was a logistic regression with diagnosis as the outcome variable, pitch variability as fixed effect and id, study and language as random effects. Our model was built using the following syntax:

glmer(Diagnosis ~ pitch_variability + (1|ID) + (1|Study) + (1|Language), family = 'binomial')

Diagnosis contained two levels; Control and Schizophrenia. The pitch variability feature was a measure of the standard deviation for each original sound recording (NB. this definition of pitch variability is different from our last assignment, where we looked at the range of pitch). We had several pitch variability measures per participant making it a repeated measures study, why random intercepts for participants were applied to meet the assumption of independence. Moreover, a separate linear model analysis of pitch variability predicted by both language and study was conducted. The

linear model analysis showed that both study and language predicted pitch variability, why they were both added as random intercepts in the final model to include the variance of study and language in the model.

The preprocessing of features (i.e. pitch-variability) consisted of scaling and mean centering the fixed effect of pitch variability. It was done after partitioning the datasets. In order to avoid data leakage from the test set to the training set while still scaling pitch variability on the same scale in both datasets, functions such as recipe(), juice() and bake() from the tidymodels package (Kuhn et al., 2020) were used. We used the bake() function to use the scaling and centering parameters of the predictors from the training set onto the test set. This ensures that pitch variability is comparable in both datasets, without polluting the test dataset. (Sigrid)

‘Initial model’ performance (GUSTAV)

The initial model was trained on the training-set and then used to predict the test-set. Performance measures of accuracy, sensitivity, specificity, PPV (positive predicted value), NPV (negative predicted value) and AUC were obtained using the caret package (Kuhn, 2020) and pROC (Robin et al., 2011).

Cross-validation and final model (ANDERS)

In order to optimize the predictive performance of the model on independent test data, the training set was divided into training- and validation sets using a 10 fold cross validation. Each fold was balanced by Diagnosis and categorized by ID, using the groupdata2 package (Olsen, 2020). This ensures a representative distribution of schizophrenics and controls, while also making sure that no participant is placed in more than one fold.

The model was then trained on 9/10 test-folds and performance measures were gathered from testing the model on the validation set (table 1). In this way we obtain several measures of the model’s performance on independent data. The beta estimates for each of the 10 models were logged. The mean of both the ten intercepts and ten slopes were then calculated and inserted to a final model whose performance was measured on the test data. The cross-validated model’s performance on the test data was then compared to the initial model’s performance. An improvement of performance of the cross-validated model was expected. (Gustav).
In other words, we have applied the method of ‘bagging’ in order to optimize the model’s predictive

performance. Bagging is a machine learning method which consists of building multiple models of the same type from different training and test sets of the original training set (Brownlee, 2016).

Results

Cross validation (SIGRID)

Table 1 shows the performance measures of the cross-validation. (Sigrid)

	Accuracy	Sensitivity	Specificity	PPV	NPV	AUC	Intercept	Slope
1	0.5310078	0	1	NaN	0.5310078	0.5	-14.37427	-0.1774191
2	0.5261324	0	1	NaN	0.5261324	0.5	-13.87192	-0.1158322
3	0.6625767	0	1	NaN	0.6625767	0.5	-13.83042	-0.1156486
4	0.5135135	0	1	NaN	0.5135135	0.5	-14.33584	-0.1176249
5	0.5204461	0	1	NaN	0.5204461	0.5	-13.88423	-0.1182842
6	0.5078534	0	1	NaN	0.5078534	0.5	-14.48317	-0.1108309
7	0.5390335	0	1	NaN	0.5390335	0.5	-13.98419	-0.0986973
8	0.5281899	0	1	NaN	0.5281899	0.5	-13.87115	-0.0945747
9	0.5201613	0	1	NaN	0.5201613	0.5	-13.90081	-0.1117938
10	0.5938865	0	1	NaN	0.5938865	0.5	-14.66594	-0.1021904

Table 1: The performance measures of each of the 10 folds using cross-validation.

Comparing initial model and final model (GUSTAV)

The estimates of initial model were as follows:
Intercept; -13.88, SE = 0.95, z score = -14,5, $p < .001$.
β: -0.11, SE = 0.49, z score = -0.25, $p > .05$ ($p = 0.8$).

The estimates of the final model were as follows:
Intercept; -14.32, SE = 0.95, z score = -15.0, $p < .001$.
β: -0.11, SE = 0.48, z score = -0.22, $p > .05$ ($p = 0.82$).

The following tables show performance metrics of the initial model and the final model.

Initial model		Reference (actual)	
		Control	Schizophrenia
Predicted	Control	358	336
	Schizophrenia	0	0

Table 2.1: Confusion matrix for the initial model

Final model		Reference (actual)	
		Control	Schizophrenia
Predicted	Control	358	336
	Schizophrenia	0	0

Table 2.2: Confusion matrix for the final model

	Initial model	Final model
Accuracy	0.52	0.52
Sensitivity	0.0	0.0
Specificity	1.0	1.0
PPV	NaN	NaN
NPV	0.52	0.52
AUC	0.5	0.5

Table 2.3: Performance measures of the initial model and final model

(Anders)

Discussion

(ANDERS) Our results showed that both our initial model as well as our final model predicted all cases to be controls. This means that both models got a specificity of 1 and a sensitivity 0, meaning that they capture all control cases but none schizophrenic (table 2.1, table 2.2). Both models had an accuracy of 0.5 which means that neither of the models performed above chance level (table 2.3). In conclusion, neither of the models had any success in predicting independent test data, and cross validation did not successfully improve predictive performance on new data of the model. Thus, pitch variability, as defined earlier, does not predict schizophrenia.

(GUSTAV) Further analysis could incorporate a feature selection algorithm (such as provided in the cvms-package) in order to assess the best combination of predictive features. However, this could lead to loss of interpretability of the model. In this analysis, the predictors were chosen on the basis of existing theory. This made sure that our predictors could be interpreted but lost predictive power. This is however always a trade-off between interpretability and predictive power.

(SIGRID) There are many other available classification processes (such as random-forest and boosted-tree). Other classification algorithms could increase the accuracy of the model. However, we are in doubt that the chosen fixed effect will ever succeed in predicting diagnosis. (Gustav)

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Portfolio 4 - Heart rate, respiration and interpersonal coordination

Anders Hjulmand, Sigrid Agersnap og Gustav Helms

Experimental Methods 3

Due 25th of November 2020

Link to code: https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_4_1

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Appendix A **20**

Introduction

(Anders): In this paper an analysis was conducted on the covariance in heart rate and respiration of participants doing a creative LEGO building task. The data was gathered by Fusaroli, R et al. (2016) and consisted of ECG and respiration measurements. Participants were paired and asked to build representations of abstract danish words e.g. *"Justice"*, *"Responsibility"*, *"Knowledge"*.
After preprocessing of the data, the final dataset used in the analysis included data from 4 studies with 6 different conditions and a total of 48 participants paired in 24 unique pairs. The data included 273.350 observations in total.
The data analysis was conducted using R (R Core Team, 2019). The models were built using the packages lme4 (Bates, Maechler, Bolker & Walker, 2015) and lmerTest (Kuznetsova, Brockhoff & Christensen, 2017).

Methods

Data and participants

(Sigrid) Data was collected from undergraduate students of Cognitive Science at Aarhus University, resulting in 97 files in total. Heart rate data (HR) and respiration (resp) data was collected with a frequency of 1000 hertz (measurements / second).
There were a total of 48 unique pairs consisting of 96 participants. Each pair completed 3-5 trials, consisting of 6 conditions: "SelfPaced", "TurnTaking", "Synchronous", "Conversation", "MovementCoop" and "MovementGuided". Conditions were shuffled across trials to keep the order randomized. There were a total of 4 studies.

Preprocessing

(Gustav) The initial data was preprocessed to prepare it for analysis. Before being merged into one data frame each data file was downsampled by taking the mean-value of every 100 rows. Outliers were removed by removing all data points that deviated by more than 2.5 standard deviations from the mean and afterwards scaled and mean-centered. All missing values were removed and the time variable was transformed into sequential data hence we were only interested in the change over time. Figure 1.1 illustrates how different steps of preprocessing changes the data. Note that the downsampled data still looks like the data before it is downsampled.

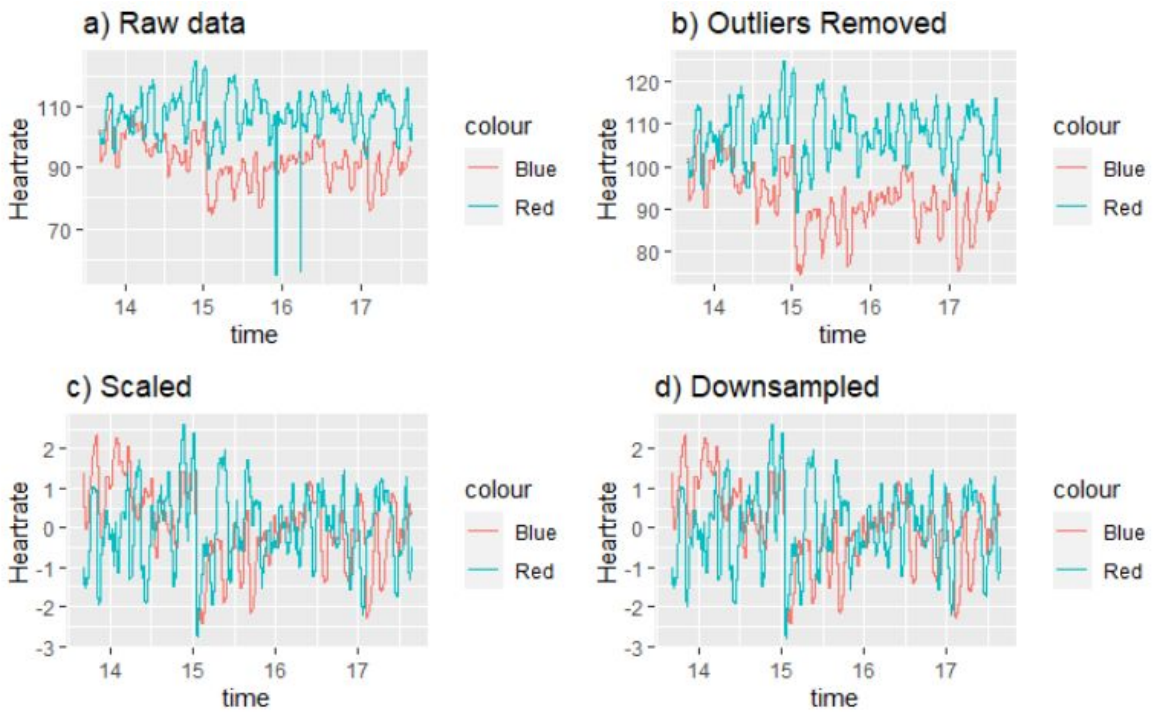

Figure 1.1: Raw data (a), outliers removed (b), scaled data (c) and downsampled data (d). Data is taken from pair1, study 1, trial 1.

Removal of artefacts

(Gustav) Even though outliers above 2.5 or below -2.5 standard deviations were removed, artifacts were still present in the data. We searched for artefacts by plotting the heart rate and the respiration data for each participant, respectively. We excluded files by eyeballing the data, trying to detect noisy data. 18 files were excluded, and the names of these can be found in table 1 in appendix A. An example of both included and excluded files for both heart rate and respiration data can be found in figure 1.2.

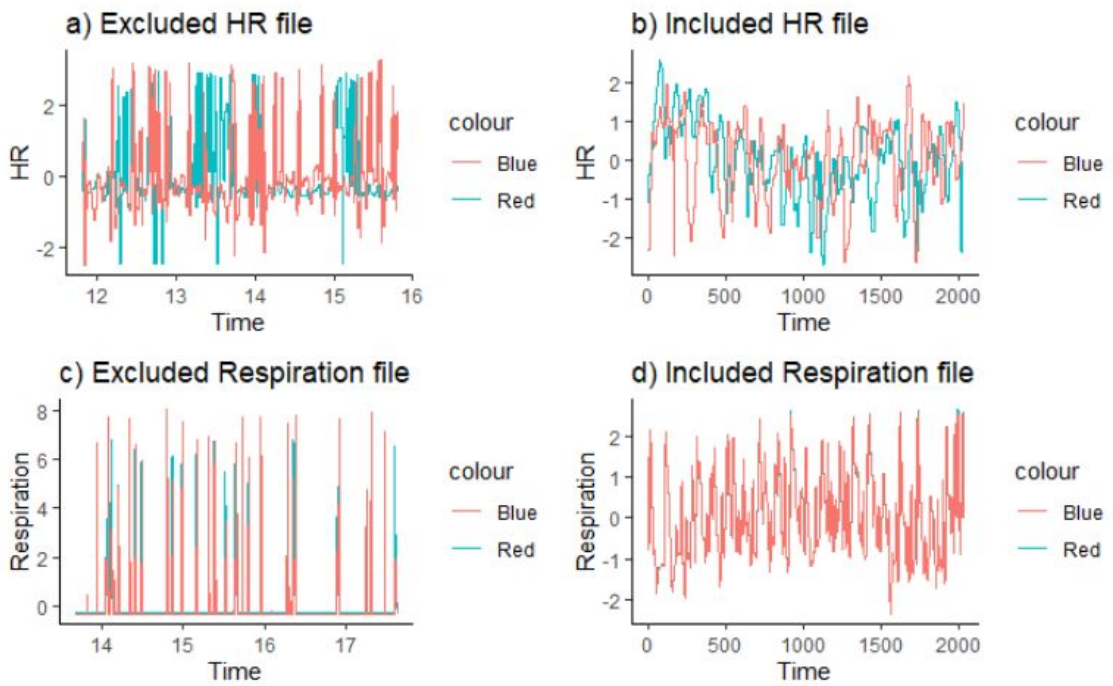

Figure 1.2: Excluded HR file (a), included HR file (b), excluded respiration file (c) and included respiration file (d).

Variables and models

(Anders) The models used in the analysis were designed to predict the change of either the respiration or HR data as a function of the first participant's data and the difference between the first participant's data and the data of the second participant. The models approximated the differential equation models proposed in Ferrer & Helm (2013). By assuming that the current change rate between each data point is equal to a differential term ($\frac{dx}{dt}$) we can use a differential equation as a model. The frequency of our measures is 10 Hz, though not instantaneous, it is a justifiable representation of the original differential term. For a more thorough review of the model design see Ferrer & Helm (2013).

The general differential equation depicted in Ferrer & Helm (2013) is written as follows:

$$\frac{dx}{dt} = a_1 * x_t + a_2(y_t - x_t) \qquad (1)$$

(Gustav) We were interested in modelling change in heart rate and respiration. We defined change as the difference between one's future heart rate (i.e. at the next time-point) and one's current heart rate. Two variables were used to predict heart rate: stability and coupling. Stability is a measure of how much the participant's heart rate is affected by her own heart rate. Coupling is a measure of how much the participant's heart rate is affected by the heart rate of the partner. These correspond to x_t and $(y_t - x_t)$ as seen in (1), respectively. After running the model we get two alpha-estimates for each condition, a_1 and a_2. Table 1.1 shows the interpretation of a_1 and a_2 given that they are significant.

	a_1	a_2
Positive	Self-instability away from homeostasis	Signal attracted to partner
Negative	Self-regulation towards homeostasis	Signal repelled from partner

Table 1.1: Interpretation of a_1 and a_2 of the model.

(Sigrid) We are interested in modelling the interactions between conditions and thereby investigate whether the different conditions elicited different tendencies of covariation. We thereby only included interaction effects in the model and left out fixed effects (thus typing ‘ : ’ rather than ‘ * ‘ in the model). None of the models included intercepts hence we were only interested in modelling the change which made the intercept redundant (the ‘0’ in the model). To meet the assumption of independence, random intercepts were added for participant, pair and study and for each of these random intercepts, a random slope for condition was added. “HR_self” was entered as a fixed effect to model stability within a participant, which corresponds to a_1. “HR_other - HR_self” was entered as a fixed effect to model coupling within the pair, which corresponds to a_2. The ‘Type’ interaction is a control condition which will be further explained in the “Control Conditions” section.

The models used in the analysis were written using the following syntax:

Heart rate model:

lmer(HR_change ~ 0 + (HR_self + (HR_other - HR_self) : Condition : Type + (0 + Condition|Participant) + (0 + Condition|Pair) + (0 + Condition|Study))

Respiration rate model:

lmer(Resp_change ~ 0 + (Resp_self + (Resp_other - Resp_self) : Condition : Type + (0 + Condition|Participant) + (0 + Condition|Pair) + (0 + Condition|Study))

The models will elicit four different outputs due to ‘Type’ being two different control conditions. This will be elaborated in the following sections.

Control conditions

(Sigrid) In order to investigate whether it is the two participants conducting the experiment who affect each other or if it is merely the condition itself, the model was run on a data set consisting of both real data and a control condition (see the 'type' predictor in the model). Thus, if the hypotheses of this experiment were to not be rejected, only the interaction effects with real data should be found to significantly predict change in heart rate. If significant results are found within both the real data and the control condition, it places mistrust in the conclusions of the significant real data. That is, the chosen hypotheses can only be supported by significant results of the real data, if the control data simultaneously produced nonsignificant results.
We created two control conditions; one with shuffled controls and one with surrogate pairs. This will result in four models total, two for each type of control condition.

Shuffled controls

(Sigrid) A new dataset was created by sampling all of the predictor variables using the sample() function in R (R Core Team, 2019). The function randomizes the values within each of the columns. That is, it does not create new values, it merely swaps the placement of the values within each column. That way a randomized dataset was created and used to predict the change in heart rate alongside the real data.

Surrogate pairs

(Gustav) The surrogate pairs were made by replacing the second participant of each pair with another second participant from another pair. Participants were only shuffled within study and condition. This made sure that the new pairs would always consist of data from the same study and condition hereby making sure that too much unaccounted variance were not added to the pairs. The new surrogate pair labels were then added to the data frame and the "HR_other" and "Resp_other" variable was updated to fit the data of the other participant in the surrogate pair. This dataset was then combined with the original data and the "Type" column was added in order to differentiate between real pairs and surrogate pairs.

Results

Plots according to condition

(Gustav) A plot depicting the heart rate data for one pair in each condition can be found in figure 2.1

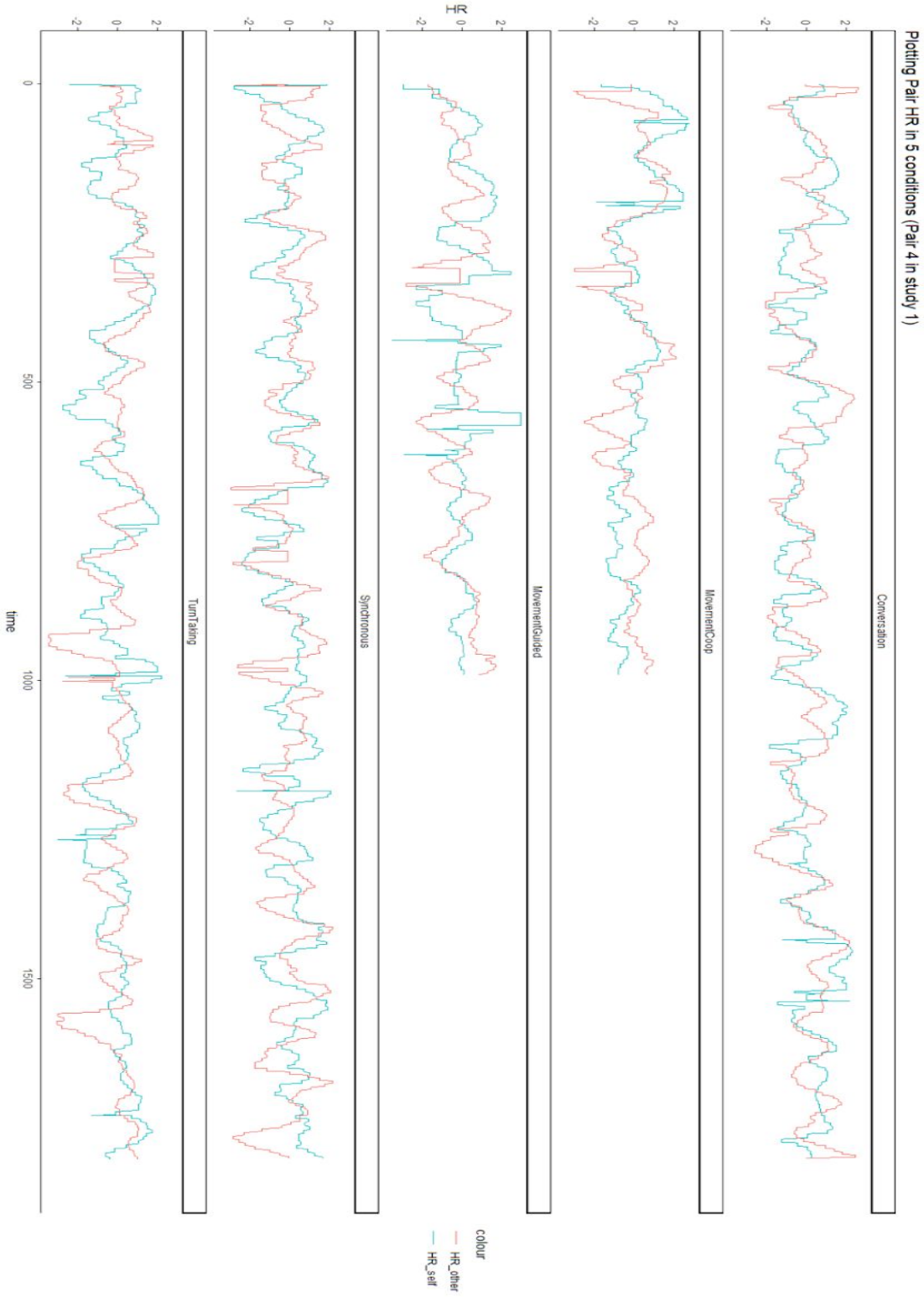

Figure 2.1: Plot depicting heart rate in the 6 different conditions for one pair

A plot depicting the respiration data for one pair in each condition can be found in figure 2.2.

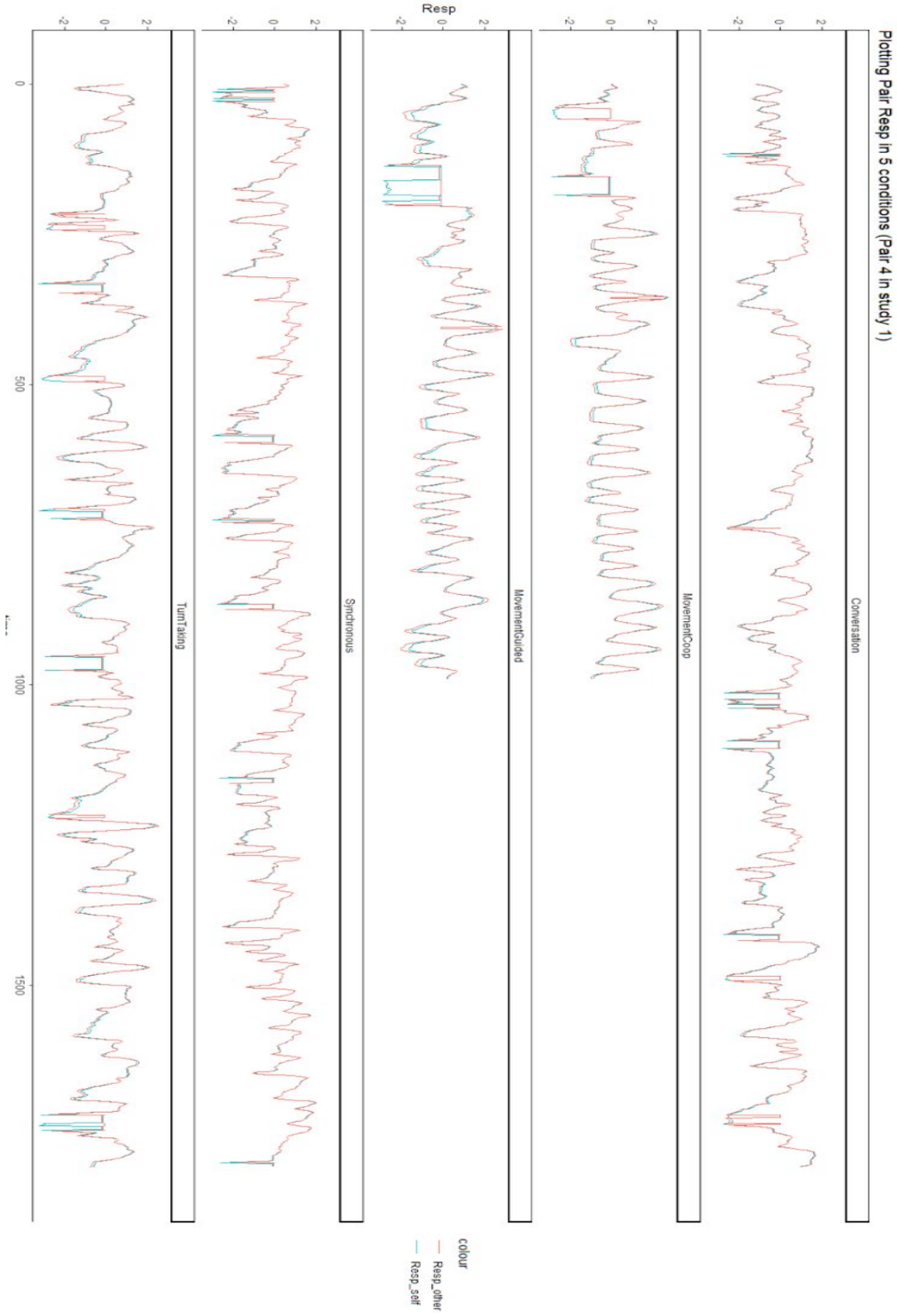

Figure 2.2: Plot depicting respiration in the 6 different conditions for one pair.

Real data and control conditions

(Anders) The tables found below display the fixed effects outputs for the four models depicted in the "Methods" section.

The marking of the p-values is as following: . = p < .1, * = p < .05, ** = p < .01, *** = p < .001

Shuffled controls for heart rate

The results of the model containing the shuffled controls for heart rate can be found in the table 2.1 below. All the below mentioned effects have 5.467e+05 degrees of freedom.

Fixed effect (interaction)	**β-estimate**	**SE**	**t-value**	**p-value**
HR_SELF real data (alpha 1)				
HR_self:Conversation:Real	-3.25e-02	9.84e-04	-33.06	<.0001 ***
HR_self:MovementCoop:Real	-3.67e-02	2.02e-03	-18.18	<.0001 ***
HR_self:MovementGuided:Real	-5.04e-02	2.00e-02	-25.15	<.0001 ***
HR_self:SelfPaced:Real	-3.47e-02	2.19e-03	15.88	<.0001 ***
HR_self:Synchronous:Real	-3.77e-02	9.28e-04	-40.64	<.0001 ***
HR_self_TurnTaking:Real	-2.85e-02	9.49-04	-30.08	<.0001 ***
HR_SELF shuffled data (alpha 1)				
HR_self:Conversation:Shuffled	-6.18e-04	9.83e-04	-0.63	0.53
HR_self:MovementCoop:Shuffled	-5.50e-03	1.97e-03	-2.791	0.0053 **
HR_self:MovementGuided:Shuffled	1.62e-03	2.00e-03	0.81	0.42
HR_self:SelfPaced:Shuffled	-4.14e-04	2.14e-03	-0.194	0.85
HR_self:Synchronous:Shuffled	-6.32e-04	9.21e-04	-0.686	0.49
HR_self:TurnTaking:Shuffled	-2.36e-04	9.44e-04	-0.250	0.80
HR_OTHER real data (alpha 2)				
HR_other:Conversation:Real	1.42e-03	9.84e-04	1.439	0.15

HR_other:MovementCoop:Real	7.27e-03	2.02e-03	3.61	0.00031***
HR_other:MovementGuided:Real	-2.959e-03	2.00e-03	-1.48	0.14
HR_other:SelfPaced:Real	7.82e-03	2.19e-03	3.58	0.00035*
HR_other:Synchronous:Real	4.78e-03	9.28e-04	5.15	<.0001***
HR_other:TurnTaking:Real	-2.87e-03	9.49e-04	-3.026	0.0025**
HR_OTHER shuffled data (alpha 2)				
HR_other:Conversation:Shuffled	1.17e-03	9.83e-04	1.190	0.23
HR_other:MovementCoop:Shuffled	-3.88e-03	1.97e-03	-1.967	0.049*
HR_other:MovementGuided:Shuffled	1.83e-03	2.00e-03	0.915	0.36
HR_other:SelfPaced:Shuffled	-7.46e-03	2.14e-03	-0.349	0.73
HR_other:Synchronous:Shuffled	2.63e-04	9.21e-04	0.286	0.77
HR_other:TurnTaking:Shuffled	2.99e-04	9.44e-04	0.316	0.75

Table 2.1: Results of both real and shuffled heart rate data. .

Shuffled controls for respiration

The results of the model containing the shuffled controls for respiration can be found in the table 2.2 below. All the below mentioned effects have 5.467e+05 degrees of freedom.

Fixed effect (interaction)	**β-estimate**	**SE**	**t-value**	**p-value**
Resp_SELF real data (alpha 1)				
Resp_self:Conversation:Real	-2.94e-01	3.71e-03	-79.46	<.0001***
Resp_self:Movement:Real	-1.95e-01	6.77e-03	-28.712	<.0001***
Resp_self:MovementGuided:Real	-2.07e-01	7.54e-03	-27.392	<.0001***
Resp_self:SelfPaced:Real	-1.67e-01	5.29e-03	-31.58	<.0001***
Resp_self:Synchronous:Real	-1.99e-01	2.83-03	-70.348	<.0001***

Resp_self:TurnTaking:Real	-2.58e-01	3.98e-03	-64.88	<.0001***
Resp_SELF shuffled data (alpha 1)				
Resp_self:Conversation:Shuffled	2.69e-04	1.03e-03	0.26	0.79
Resp_self:Movement:Shuffled	4.76e-03	2.07e-03	2.30	0.0213*
Resp_self:MovementGuided:Shuffled	-5.13e-04	2.10e-03	-0.25	0.81
Resp_self:SelfPaced:Shuffled	-5.59e-03	2.24e-03	-2.50	0.0126*
Resp_self:Synchronous:Shuffled	-9.53e-04	0.65e-04	-0.99	0.32
Resp_self:TurnTaking:Shuffled	1.86-04	9.89e-04	0.19	0.85
Resp_OTHER real data (alpha 2)				
Resp_other:Conversation:Real	2.65e-01	3.71-03	71.62	<.0001***
Resp_other:MovementCoop:Real	1.63e-01	6.77e-03	23.99	<.0001***
Resp_other:MovementGuided:Real	1.75e-01	7.54e-03	23.21	<.0001***
Resp_other:SelfPaced:Real	1.36e-01	5.29e-03	25.64	<.0001***
Resp_other:Synchronous:Real	1.69e-01	2.83e-03	59.84	<.0001***
Resp_other:TurnTaking:Real	2.31e-01	3.98e-03	58.20	<.0001***
Resp_OTHER shuffled data (alpha 2)				
Resp_other:Conversation:Shuffled	6.33e-04	1.03e-03	0.62	0.54
Resp_other:MovementCoop:Shuffled	1.36e-03	2.07e-03	0.66	0.51
Resp_other:MovementGuided:Shuffled	-3.44e-03	2.10e-03	-1.64	0.10
Resp_other:SelfPaced:Shuffled	2.11e-03	2.24e-03	0.94	0.35
Resp_other:Synchronous:Shuffled	-7.14e-04	9.65e-04	-0.74	0.46
Resp_other:TurnTaking:Shuffled	-1.79e-04	9.89e-04	-0.18	0.86

Table 2.2: Results of both real and shuffled respiration data.

Surrogate pair controls for heart rate

The results of the model containing the surrogate pair controls for heart rate can be found in the table 2.3 below. All the below mentioned effects have 4.9e+05 degrees of freedom.

Fixed effect (interaction)	**β-estimate**	**SE**	**t-value**	**p-value**
HR_SELF real data (alpha 1)				
HR_self:Conversation:Real	-3.52e-02	9.90e-04	-32.82	<.0001 ***
HR_self:MovementCoop:Real	-3.67e-02	2.03e-03	-18.07	<.0001 ***
HR_self:MovementGuided:Real	-5.04e-02	2.02e-03	-24.99	<.0001 ***
HR_self:SelfPaced:Real	-3.47e-02	2.20e-03	-15.78	<.0001 ***
HR_self:Synchronous:Real	-3.77e-02	9.34e-04	-40.39	<.0001 ***
HR_self_TurnTaking:Real	-2.85e-02	9.54e-04	-29.90	<.0001 ***
HR_SELF surrogate pairs (alpha 1)				
HR_self:Conversation:Surrogate	-3.41e-02	1.06e-03	-32.11	<.0001 ***
HR_self:MovementCoop:Surrogate	-3.47e-0.2	2.13e-03	-16.31	<.0001 ***
HR_self:MovementGuided:Surrogate	-4.77e-0.2	2.13e-03	-22.41	<.0001 ***
HR_self:SelfPaced:Surrogate	-3.44e-02	2.24e-03	-15.33	<.0001 ***
HR_self:Synchronous:Surrogate	-3.94e-02	1.02e-03	-38.54	<.0001 ***
HR_self:TurnTaking:Surrogate	-3.04e-02	1.14e-03	-26.64	<.0001 ***
HR_OTHER real data (alpha 2)				
HR_other:Conversation:Real	1.42e-03	9.90e-04	1.43	0.15
HR_other:MovementCoop:Real	7.27e-03	2.03e-03	3.58	0.003 ***
HR_other:MovementGuided:Real	-2.96e-03	2.02e-03	-1.47	0.14
HR_other:SelfPaced:Real	7.82e-03	2.20e-03	3.55	0.0003 ***
HR_other:Synchronous:Real	4.78e-03	9.34e-04	5.12	<.0001 ***

HR_other:TurnTaking:Real	-2.87e-03	9.54e-04	-3.01	0.0026 **
HR_OTHER surrogate pair (alpha 2)				
HR_other:Conversation:Surrogate	1.22e-03	1.06e-03	1.15	0.25
HR_other:MovementCoop:Surrogate	2.82e-03	2.13e-03	1.33	0.18
HR_other:MovementGuided:Surrogate	9.02e-04	2.13e-03	0.42	0.67
HR_other:SelfPaced:Surrogate	1.15e-03	2.24e-03	0.51	0.61
HR_other:Synchronous:Surrogate	1.52e-03	1.02e-03	1.49	0.14
HR_other:TurnTaking:Surrogate	-5.31e-04	1.14e-03	-0.47	0.64

Table 2.3: Results of both real and surrogate pairs heart rate data.

Surrogate pair controls for respiration

The results of the model containing the surrogate pair controls for respiration can be found in the table 2.4 below. All the below mentioned effects have 4.948e+05 degrees of freedom.

Fixed effect (interaction)	**β-estimate**	**SE**	**t-value**	**p-value**
Resp_SELF real data (alpha 1)				
Resp_self:Conversation:Real	-2.94e-01	3.65e-03	-80.60	<.0001***
Resp_self:MovementCoop:Real	-1.95e-01	6.68e-03	-29.12	<.0001***
Resp_self:MovementGuided:Real	-2.06e-01	7.43e-03	-27.78	<.0001***
Resp_self:SelfPaced:Real	-1.67e-01	5.22e-03	-32.02	<.0001***
Resp_self:Synchronous:Real	-1.99e-01	2.79e-03	-71.35	<.0001***
Resp_self_TurnTaking:Real	-2.58e-01	3.92e-03	-65.80	<.0001***
Resp_SELF surrogate data (alpha 1)				
Resp_self:Conversation:Surrogate	-3.71e-02	1-08e-03	-34.12	<.0001***

Resp_self:MovementCoop:Surrogate	-3.36e-02	2.17e-03	-15.50	<.0001***
Resp_self:MovementGuided:Surrogate	-3.76e-02	2.22e-03	-16.94	<.0001***
Resp_self:SelfPaced:Surrogate	-4-57e-02	2.37e-03	-19.26	<.0001***
Resp_self:Synchronous:Surrogate	-3.83e-02	1.05e-03	-36.38	<.0001***
Resp_self:TurnTaking:Surrogate	-3.48e-02	1.17e-03	-29.7	<.0001***
Resp_OTHER real data (alpha 2)				
Resp_other:Conversation:Real	2.65e-01	3.65e-03	72.64	<.0001***
Resp_other:MovementCoop:Real	1.63e-01	6.67e-03	24.33	<.0001***
Resp_other:MovementGuided:Real	1.75e-01	7.43e-03	23.53	<.0001***
Resp_other:SelfPaced:Real	1.40e-01	5.21e-03	26.00	<.0001***
Resp_other:Synchronous:Real	1.69e-01	2.78e-03	60.69	<.0001***
Resp_other:TurnTaking:Real	2.314e-01	3.92e-03	59.03	<.0001***
Resp_OTHER surrogate data (alpha 2)				
Resp_other:Conversation:Surrogate	6.16e-04	1.09e-03	0.57	0.57
Resp_other:MovementCoop:Surrogate	-7.36e-04	2.17e-03	-0.34	0.73
Resp_other:MovementGuided:Surrogate	-3.95e-04	2.21e-03	-1.78	0.075 .
Resp_other:SelfPaced:Surrogate	1.11e-03	2.37e-03	0.47	0.63
Resp_other:Synchronous:Surrogate	2.68e-03	1.05e-03	2.54	<0.5*
Resp_other:TurnTaking:Surrogate	5.36e-04	1-17e-03	0.46	0.647

Table 2.4: Results of both real and surrogate pairs for the respiration data.

Discussion

Two different control conditions serve two different purposes

(Gustav) Two control conditions were used in the analysis: shuffled controls and surrogate pairs. In the data where shuffled controls were used, all the data for HR_self, HR_other, Resp_self and Resp_other have been randomly shuffled. In the following discussion, the shuffled control condition is used to validate or invalidate the significant results of only HR_self and Resp_Self (a_1).
The other control condition, surrogate pairs, assigns a new pair to each participant and thus changes the values HR_other and Resp_other while keeping the values of HR_self and Resp_self constant. Thus, the surrogate pair control condition is used to validate or invalidate the significant results for only HR_other and Resp_other (a_2).
We could also have chosen to use shuffled controls as an extra control condition for HR_other and Resp_other. However, we think that the surrogate pairs condition is more controlled (conforms to study and condition) and thereby better suited as a control condition for HR_other and Resp_other.

Heart rate

Shuffled Controls

(Anders) HR_self was found to be significant across all conditions in the real data (table 2.1). This indicates a stability within each individual's heart rate. However, HR_self was also significant in the shuffled data for the condition “MovementCoop”. This thereby invalidates the significant result for HR_self in the same condition. In the remaining 5 conditions, all alpha-estimates are negative, meaning that participants are moving toward homeostasis. Thus, using shuffled controls as a control condition, we find a significant effect of stability in 5 out of 6 of the conditions (excluding “MovementCoop”).

Surrogate Pairs

(Anders) All the results for the real data for HR_self are the same as mentioned above in the “Shuffled controls”, in terms of significance and alpha estimates. In the surrogate pair data we also find significant results across all conditions for HR_self (table 2.3). This does not invalidate the significant results for HR_self in the real data, because the surrogate data only changes the values of HR_other, not HR_self.

We can use surrogate pairs to validate or invalidate HR_other.
HR_other was found to be significant in 4 out of 6 conditions in the real data. The significant conditions were: “MovementCoop”, “SelfPaced”, “Synchronous” and “Turntaking”.

HR_other was not found to be significant in any of the conditions in the surrogate pair data. Thus, using surrogate pairs as a control condition for HR_other, we find that it is significant across all 4 conditions. In the 4 conditions the alpha values were positive for “MovementCoop”, “SelfPaced” and “Synchronous” while being negative for “Turntaking”. Thus, we find a significant effect of coupling *towards* the partner in the conditions “MovementCoop”, “Selfpaced” and “Synchronous”. In contrast, there is a significant effect of coupling *away* from the partner in “TurnTaking”.

Respiration

Shuffled Controls

(Sigrid) The real data for Resp_self were in general found to be significant, while the shuffled data were not, apart from two instances of Resp_self being significant even though the data were shuffled (table 2.2). These two instances were, however, significant at a higher threshold (p = <. 05) than the real data (p = <.0001). Though these two instances could be due to chance, we cannot say anything with certainty. This discredits the significant results of Resp_self in the two conditions in question, which are ‘MovementCooperated’ and ‘SelfPaced’.
If we set aside the two two incidents of significant results of the shuffled data, the results in general indicate that the respiration of the participant in question seems to remain stable, that is, it seems to go towards homeostasis because of the negative alpha values.

Surrogate Pairs

(Sigrid) As mentioned, it is only the effect of Resp_other (a_2) that is relevant in the surrogate pair control condition. All but one of the effects of the surrogate pair data were found to be non-significant, while all of the real pairs were found to be significant (table 2.4). The only significant surrogate pair effect was found in the ‘Synchronous’ condition (p = <. 05). Although the effects of the real data were significant at a lower level (p = <.0001), the significance of the surrogate pair effect still discredits the significant results of Resp_other in the mentioned condition in the real data.
The effects of Resp_other in the remaining conditions for the real data all indicate that the respiration of the participant in question aligns with the respiration of the other participant.

Summary of interpreted results

The following table provides a summary for all the significant results (3.1).

		Significant conditions	Alpha direction	Effect sizes
Heart Rate	HR_self stability (a_1)	Conversation	stability	-3.25e-02
		Movement Guided	stability	-5.04e-02
		Self Paced	stability	-3.47e-02
		Synchronous	stability	-3.77e-02
		Turn Taking	stability	-2.85e-02
	HR_other coupling (a_2)	Movement Coop	coupling	7.27e-03
		Self Paced	coupling	7.82e-03
		Synchronous	coupling	4.78e-03
		Turn Taking	**decoupling**	-2.87e-03
Respiration	Resp_self stability (a_1)	Conversation	stability	-2.94e-01
		Movement Guided	stability	-2.07e-01
		Synchronous	stability	-1.99e-01
		Turn Taking	stability	-2.58e-01
	Resp_other coupling (a_2)	Conversation	coupling	2.65e-01
		Movement Coop	coupling	1.63e-01
		Movement Guided	coupling	1.75e-01
		Self Paced	coupling	1.36e-01

		Turn Taking	coupling	2.31e-01

Table 3.1: Summary of the significant scaled effects size for HR and respiration data across all significant conditions.

(Anders) We found an effect of stability in both heart rate and respiration in the following 4 conditions: Conversation, Movement Guided, Synchronous and Turn Taking.
We found an effect of coupling within pairs in both heart rate and respiration in two conditions: Movement Coop and Self paced. The condition "Turn Taking" produced significant results for both heart rate and respiration. However, in the respiration data it was found to produce coupling, whereas in the heart rate data it was found to produce a decoupling effect. This is a contradictory result.

In general, it seems that the scaled effect sizes are larger for the respiration data compared to the heart rate data. This will be elaborated in the last section of this assignment.

(Sigrid) The largest scaled effects of the respiration data are observed in the 'Conversation' and the 'TurnTaking' condition for both Resp_self and Resp_other. In Resp_self the difference between the largest and the smallest scaled effect size is 0.01, while in Resp_other the difference between the largest and the smallest effect size is 0.13. Thus, the difference between the scaled effect sizes of Resp_self is fairly small. That is, the four significant conditions do have an effect on the stability of the participant's respiration, however, the effect does not vary a lot within the four conditions.
The variation of the effect sizes for Resp_other does, however, seem to vary a bit more when compared to Resp_self. A variation of 0.13 between effect sizes could be concluded to be a relatively big effect, since the effect sizes in general are quite small.

The variance of the scaled effect sizes for stability and coupling in heart rate is relatively small. The conditions yield effect sizes that are quite similar to each other. Thus, there is a low variability of effect sizes within conditions.

Is respiration coordination a likely driver of heart coordination?

(Anders) The analysis showed a significant covariation effect for both respiration and heart rate between participants within pairs. This initially leads to the question of whether respiration and heart rate internally are correlated and if so, whether participants in reality only align to each other on one of the two parameters. In that case, the other parameter shows a significant covariation effect as a consequence to its correlation to the aligned parameter. The next sections will be discussing the causation of the significant effects found in our analysis.

(Sigrid) Our results showed that the covariation effect of respiration between participants was stronger than the covariation effect of heart rate (see figure 1.2b and 1.2d). If heart rate is dependent on respiration, it could possibly be explained by the fact that participants only aligned their respiration to the respiration of their partner and not their heart rate. However, since heart rate is correlated with respiration, the heart rate of the two participants, respectively, becomes synchronous as a consequence of this. This could be tested by introducing two new hypotheses; H1) Heart rate is predicted by respiration, and H2) Participants align their respiration to the respiration of their partner, but not their heart rate.

(Gustav) To test H1, one could simply run a model that tested whether heart rate was predicted by respiration. If respiration significantly predicts heart rate, it could be argued that since the covariation effect is much more apparent in the respiration data than in the heart rate data, the covariation effect found in heart rate data is a secondary effect of respiration rather than an alignment to the partner's heart rate. A possible explanation to this could be that respiration is a process that is easy to detect externally both visually and auditorily in bodily motions and sounds associated with respiration. Conversely, it is much more difficult to detect heart rate only by visual and auditory cues. However, Li, Chen, Zhao & Pietikäinen (2014) have been able to successfully detect heart rate only by visual cues from video data of faces. If a machine is able to detect heart rate only by visual cues why should humans not be able to do it as well? In order to determine the two parameters' true covariation effect a new study design is needed that can make an accurate dissociation between respiration and heart rate covariation. The question is if this is possible since it is hard to imagine two tests that separate respiration and heart rate since they will always be interconnected in the human system.

(Gustav) The results of our study show that a covariation effect is present. Participants did align both their respiration and heart rate to that of their partner. However, we are not able to determine the internal relationship between heart rate and respiration. We are also not able to determine whether the covariation effect found for heart rate and respiration are a true effect or a secondary effect of the alignment to the other parameter i.e. we cannot determine the dependency of the two parameters.

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Appendix A

The following 18 files were removed due to artefacts in either the heart rate and/or the respiration data. Each row corresponds to one file.

Study	Pair	Trial
1	1	1
1	1	2
1	3	2
2	5	1
2	5	2
2	5	3
2	6	1
2	6	2
2	6	3
2	8	1
2	8	2
2	8	3
3	1	3
3	9	2
4	1	4
4	3	3
4	3	5
4	6	1

Table 1: The 18 files which were excluded due to artefacts. Each row corresponds to one file.

Portfolio 5 - Meta analysis of voice features in schizophrenia

Anders Hjulmand, Sigrid Agersnap og Gustav Helms

Experimental Methods 3

Due 11th of December 2020

Link to code: https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_5

Introduction

(Anders) This meta-analysis-paper seeks to compare different studies' results on predicting schizophrenia by the vocal feature of pitch.

Methods

(Sigrid) The full process of the literature review can be found in the PRISMA-chart in appendix A. For the quantitative synthesis, 48 articles were included. Some articles included several studies, so in total there were 50 studies. Out of these 50 studies, we conducted meta-analysis on 6 of the studies. These 6 studies all included quantitative measures for mean and standard deviation of pitch for both groups; healthy controls (HC) and schizophrenics (SZ). Many other variables than pitch have been proposed to diagnose schizophrenia. This meta-study focuses only on pitch as a predicting variable for schizophrenia.The 6 studies used for the meta analysis only represented (6/50 = 0.12) 12% of the overall studies reviewed.

The meta-analysis was conducted using the metafor package (Viechtbauer, 2010) in R (R Core Studio, 2019).

Variables

(Gustav) The mean and standard deviation of pitch were assessed for both the control and schizophrenic group across all studies. A standardized effect size and the associated variance was calculated by use of the escalc() function from the metafor package (Viechtbauer, 2010).

Analysis and models

(Anders) The analysis was conducted on both the original dataset consisting of six studies, and on a dataset including the studies from assignment 3. These datasets will from here on be referred to as *the original dataset* and *the full dataset*. The data from assignment 3 consists of 4 studies, which we have pooled into one study.

The models were made using the rma() function from the metafor package (Viechtbauer, 2010). The rma() function creates a version of a general linear mixed effects model, which is suitable for meta analyses. The model only consisting of random effects was written using the following syntax:

$$m_{random} <- rma(yi, vi, data, slab = StudyID)$$

An additional model with fixed effects was made. The model was built using the following syntax:

$$m_{fixed} <- rma(yi, vi, mods = cbind(TYPE\ OF\ TASK), data, slab = StudyID)$$

Forest-plots, bajaut-plots and funnel-plots were made using the functions forest(), bajaut() and funnel(), respectively, from the metafor-package (Viechtbauer, 2010). The functions regtest() and ranktest() were used to calculate objective measures for funnel-plot asymmetry, also from the metafor-package.

Results

Results from the fixed effect model

(Sigrid) Our fixed effect model on the original dataset showed no significant effect of the moderator ‘type of task’ with QM(df = 1) = 0.5154, p-val = 0.4728. This was also the case for the dataset where study 3 had been added with QM(df = 1) = 0.5952, p-val = 0.4404. The analysis thereby indicates that the type of tasks condition does not account for any of the heterogeneity in the data.

Results from the original dataset

(Gustav) All the following results are from the random effects model. The meta-analysis on the original dataset indicates that the studies were not heterogeneous, Q(df = 5) = 9.8472, p-val = 0.0797. The Tau^2 statistic is also indicative of no heterogeneity, tau^2 = 0.07 (SE = 0.0908). The I^2 statistic was 50.29% which represents a moderate variance between the studies. The summary effect size was non-significant at -0.16 CI[-0.46,0.14], SE = 0.15, $p > 0.05$ (p = 0.29). The forest plot can be found in figure 1.

Egger's regression test indicated funnel-plot asymmetry, z = -1.58, p = 0.11. The funnel plot can be found in figure 2. Kendall's rank correlation test also indicated funnel-plot asymmetry, $\tau = -0.20$, p = 0.72.

Since the studies were found to be relatively homogenous and the full dataset only contains 6 studies, we will not remove the most influential studies found in the upper right quadrant of the Baujat-plot (Figure 2).

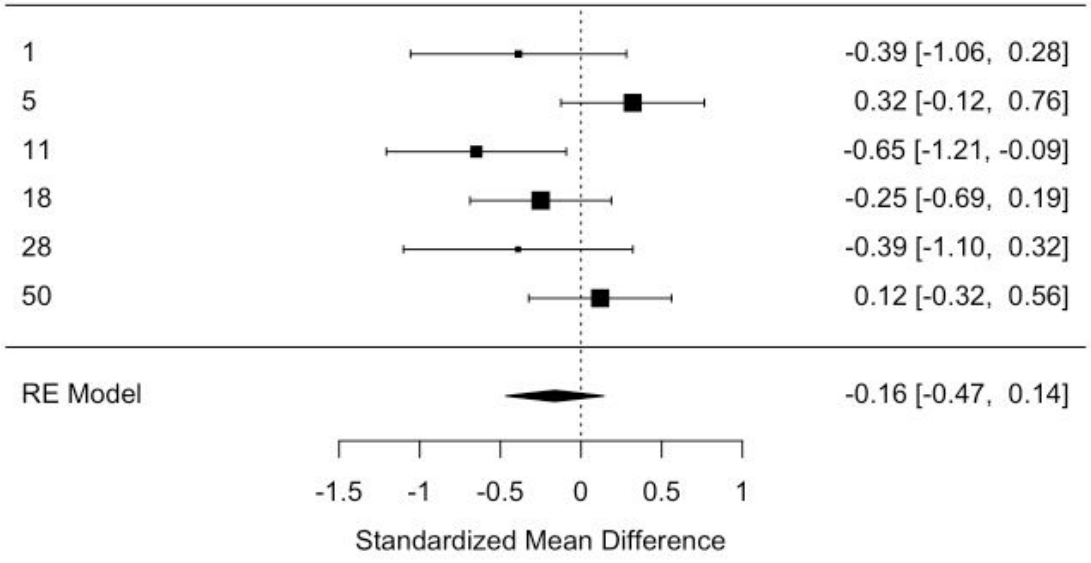

Figure 1 Forest plot for the original dataset

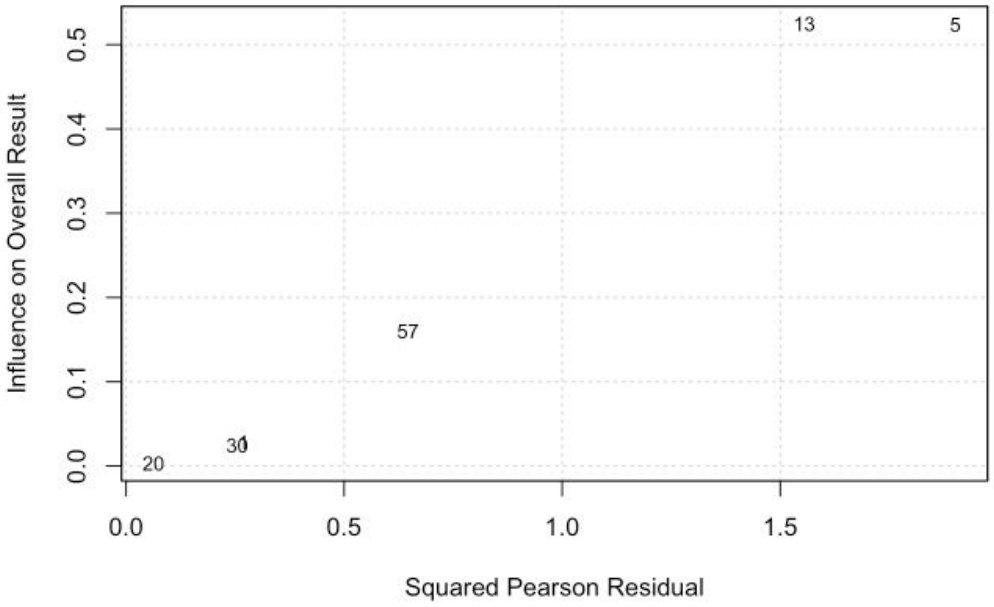

Figure 2: Baujaut plot for the original dataset

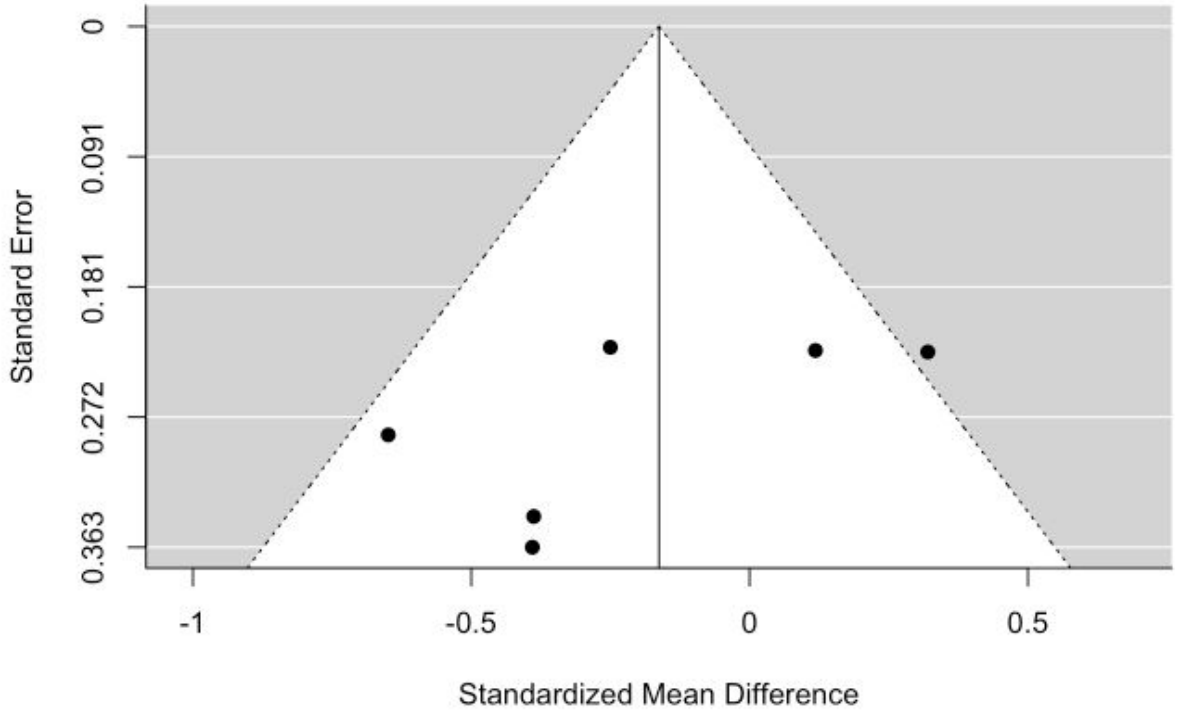

Figure 3: Funnel plot of the original dataset

Results from full dataset

(Anders) All the following results are from the random effects model made from the full dataset.

The output of the random effects model indicates that the studies were not heterogeneous, Q (df = 6) = 9.96, p = 0.13. The amount of variance which could be attributed to heterogeneity between studies rather than within studies was found to be low to moderate, I^2 = 41.42 %. The total amount of heterogeneity was calculated using tau^2, tau^2 = 0.04, SE = 0.05. The summary effects size was non-significant at -0.16, CI[-0.38, 0.07], SE = 0.11, p > 0.05 (p = 0.17).

Egger's regression test showed funnel-plot asymmetry, z = -0.81, p > 0.05 (p = 0.42).

Since the studies were found to be relatively homogenous and the full dataset only contains 7 studies, we will not remove the most influential studies found in the upper right quadrant of the Baujat-plot (Figure 5).
Kendall's tau rank test also showed funnel-plot asymmetry, τ = -0.24, p > 0.05 (p = 0.56).

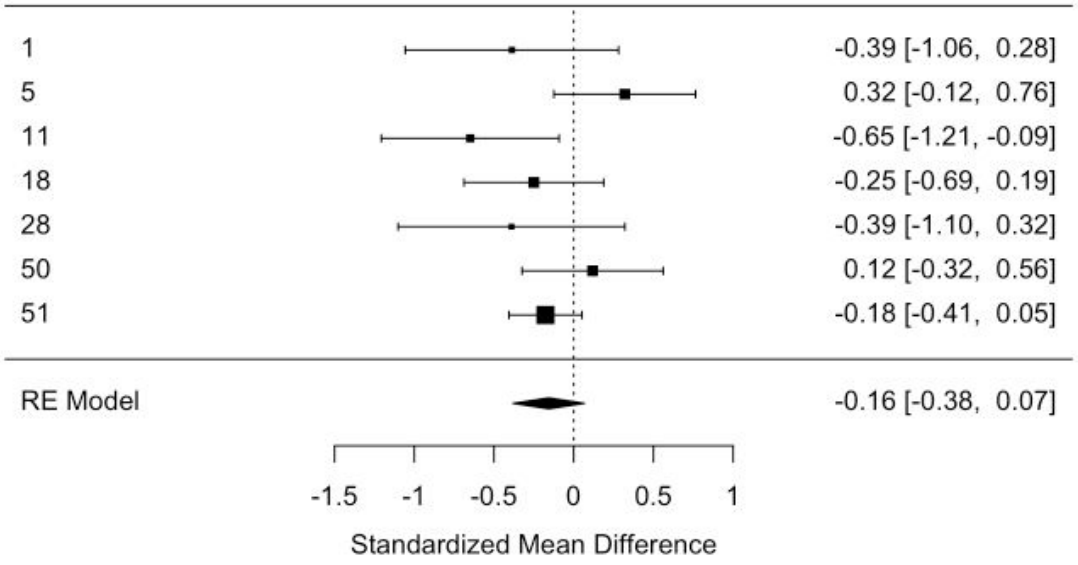

Figure 4: Forest plot for the full dataset. Study 51 indicates the newly added study

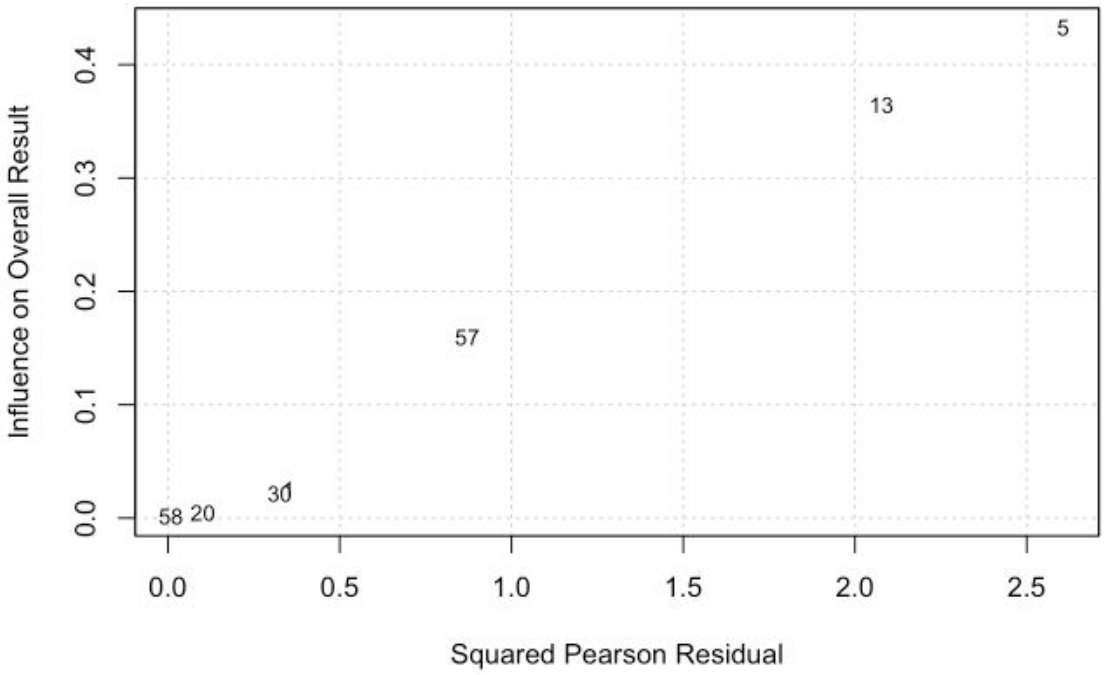

Figure 5: Baujat plot for the full dataset.

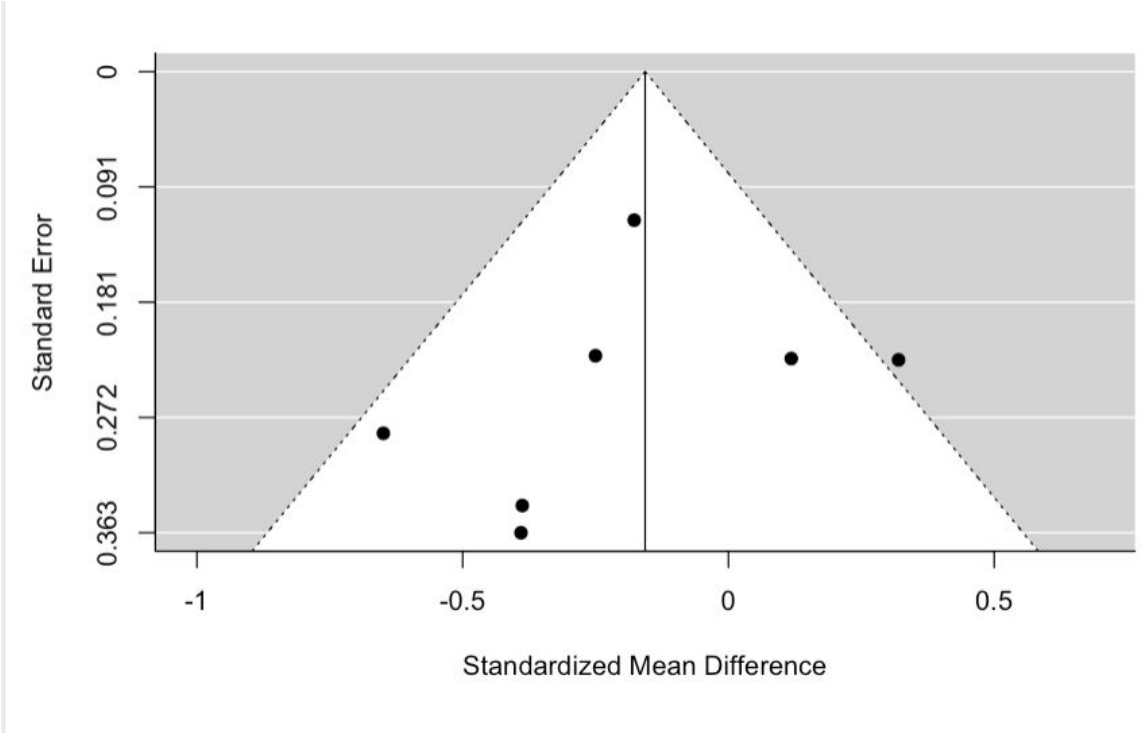

Figure 6: Funnel plot of the full dataset.

Discussion

(Sigrid) The results from the original dataset suggested that the studies were not heterogenous. Thus, the studies were eligible to conduct a meta-analysis. The studies summary effect size was calculated and found to be non-significant. The tests investigating funnel-plot asymmetry found that there might be a publication bias, since the funnel-plots were significantly asymmetrical.

The results from the full dataset showed that the studies were not heterogenous. The summary effect size was, however, not significant, thereby indicating that there is no common effect size of the studies. The tests investigating funnel-plot asymmetry found that there might be a publication bias, since the funnel-plots were significantly asymmetrical.

Although, the numerical tests for publication bias for both the original and the full dataset showed funnel-plot asymmetry, we question the validity of these tests. The funnel plots only include respectively 6 and 7 studies which makes it difficult to determine anything about the distribution of the studies in the funnel plot. Likewise, it is not possible to determine whether a distribution is gaussian with only 6 or 7 data points. Therefore, we conclude that it is not possible to determine whether there is a publication bias in the data because of the low number of studies included in the analysis.

(Gustav) Adding study 3 to the original set (the full dataset) improved the results of the analysis. Tau^2 dropped from 0.07 to 0.04 indicating that the total amount of heterogeneity in the dataset decreased after adding study 3. Also, the I^2 decreased from 50.29 % to 41.42 %, indicating that less of the variance in the data was caused by heterogeneity, but rather by with-in study variance. Adding study 3 thereby resulted in less heterogeneity in the meta-analysis.

The summary effect size was still insignificant after adding study 3. However, the p-value did improve (from $p = 0.29$ to $p = 0.17$), indicating that the summary effect size was closer to being marginally significant. Thus, adding study 3 improved the non-heterogeneity and the p-value of the summary effect size. We propose that adding more studies to the meta-analysis might result in a significant summary effect size and a more accurate estimate of publication bias. It seems that our meta-analysis lacks power because of the low sample size, but there might still be a true underlying effect.

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Appendix A

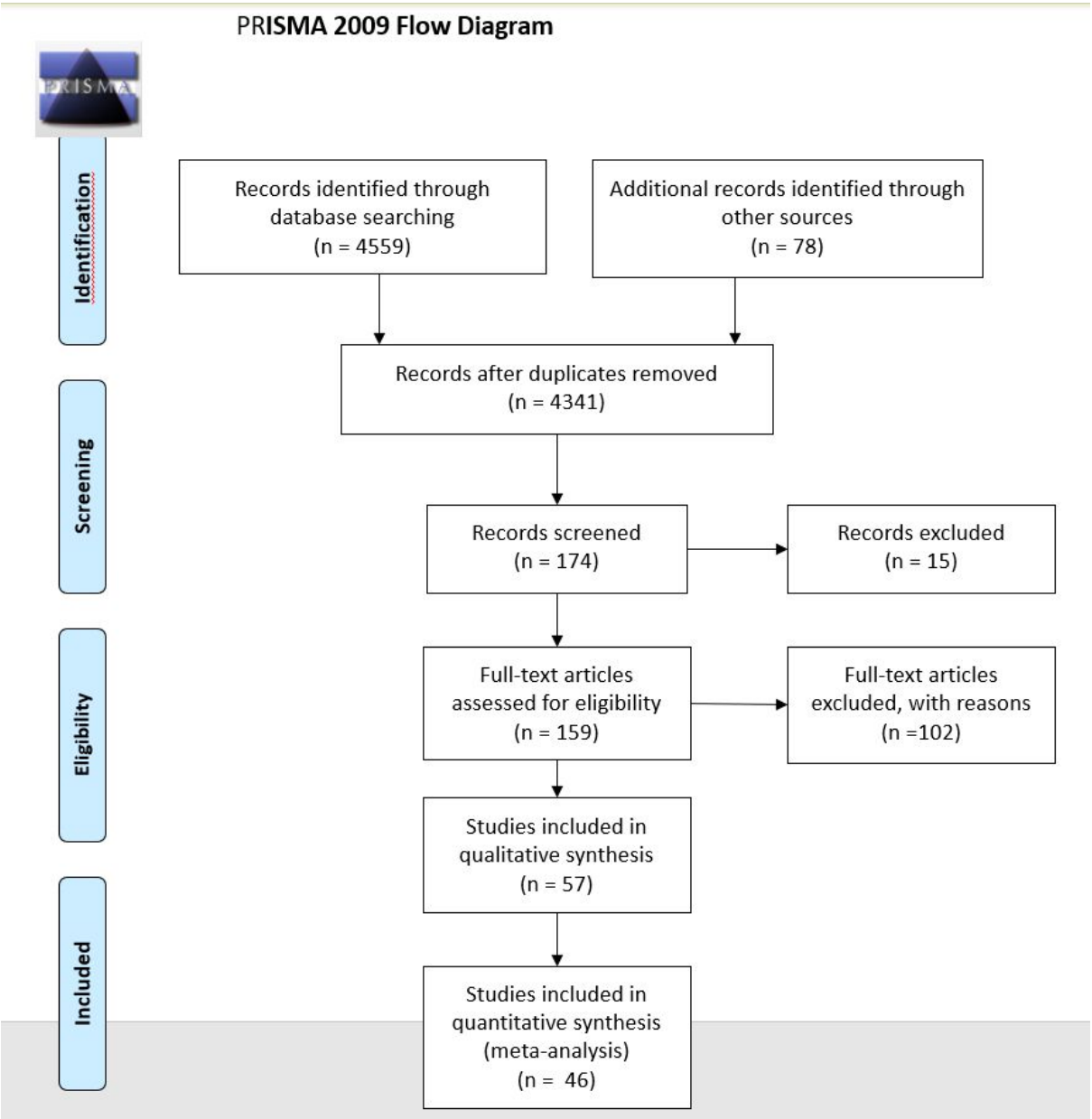

Figure A: PRISMA-chart depicting the literature-search process.